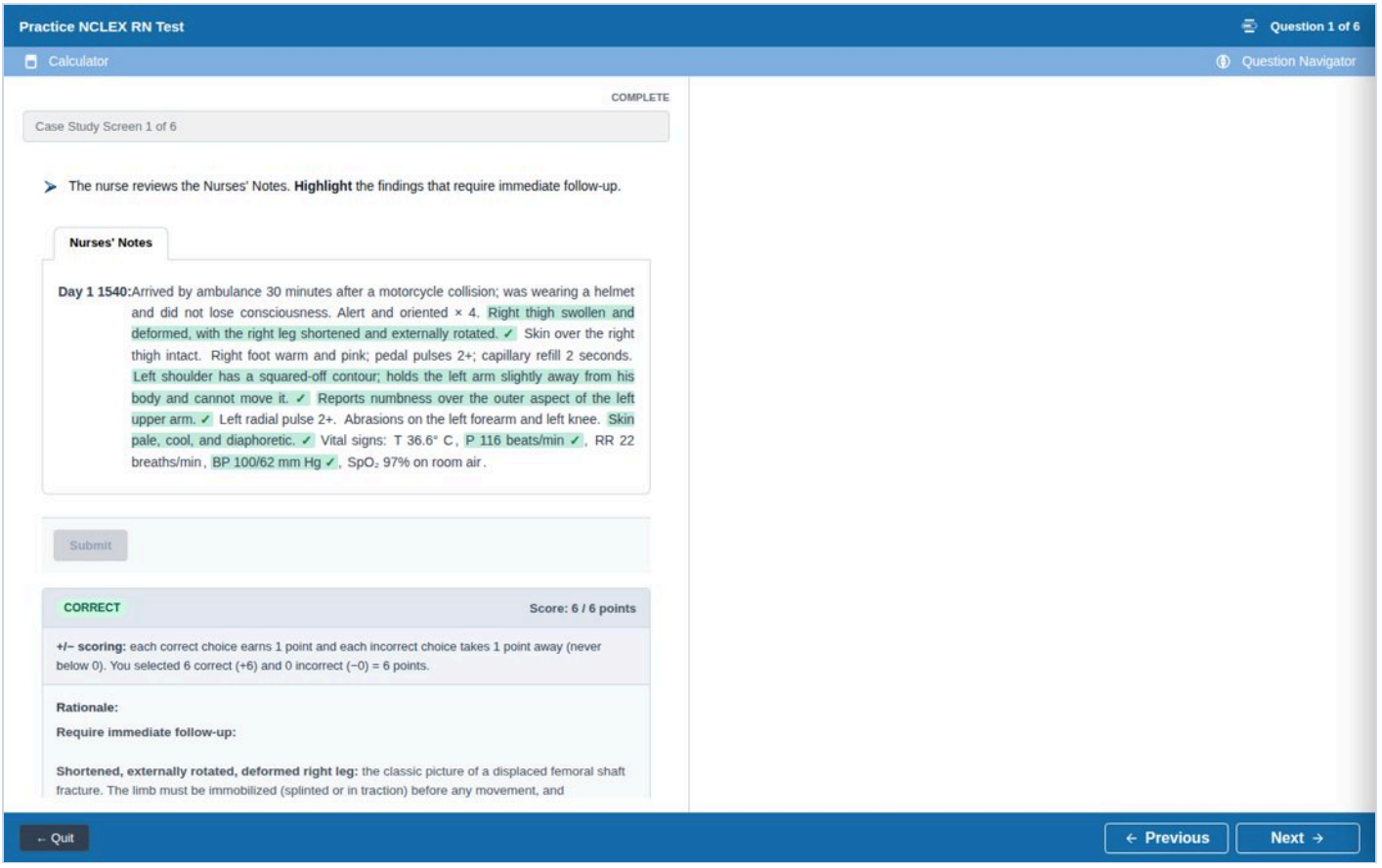
NURS 1017 Unit 6 Case Studies 2 to 7 (review copy)

For each screen: the question with the answer key filled in and submitted, then the full rationale as students see it after Submit. All 36 screens score full marks with their answer keys and all six cases pass the readiness check.

- Case Study 2: Fracture, dislocation, fat embolism and bone healing
- Case Study 3: Osteomyelitis
- Case Study 4: Osteoporosis and vertebral compression fracture
- Case Study 5: Myasthenia gravis
- Case Study 6: Osteoarthritis
- Case Study 7: Fibromyalgia

Case Study 2 (Fracture, dislocation, fat embolism and bone healing), screen 1 of 6: Recognize cues



Case Study 2, screen 1: rationale

CORRECT **Score: 6 / 6 points**

+/− scoring: each correct choice earns 1 point and each incorrect choice takes 1 point away (never below 0). You selected 6 correct (+6) and 0 incorrect (−0) = 6 points.

Rationale:

Require immediate follow-up:

Shortened, externally rotated, deformed right leg: the classic picture of a displaced femoral shaft fracture. The limb must be immobilized (splinted or in traction) before any movement, and neurovascular status is checked often.

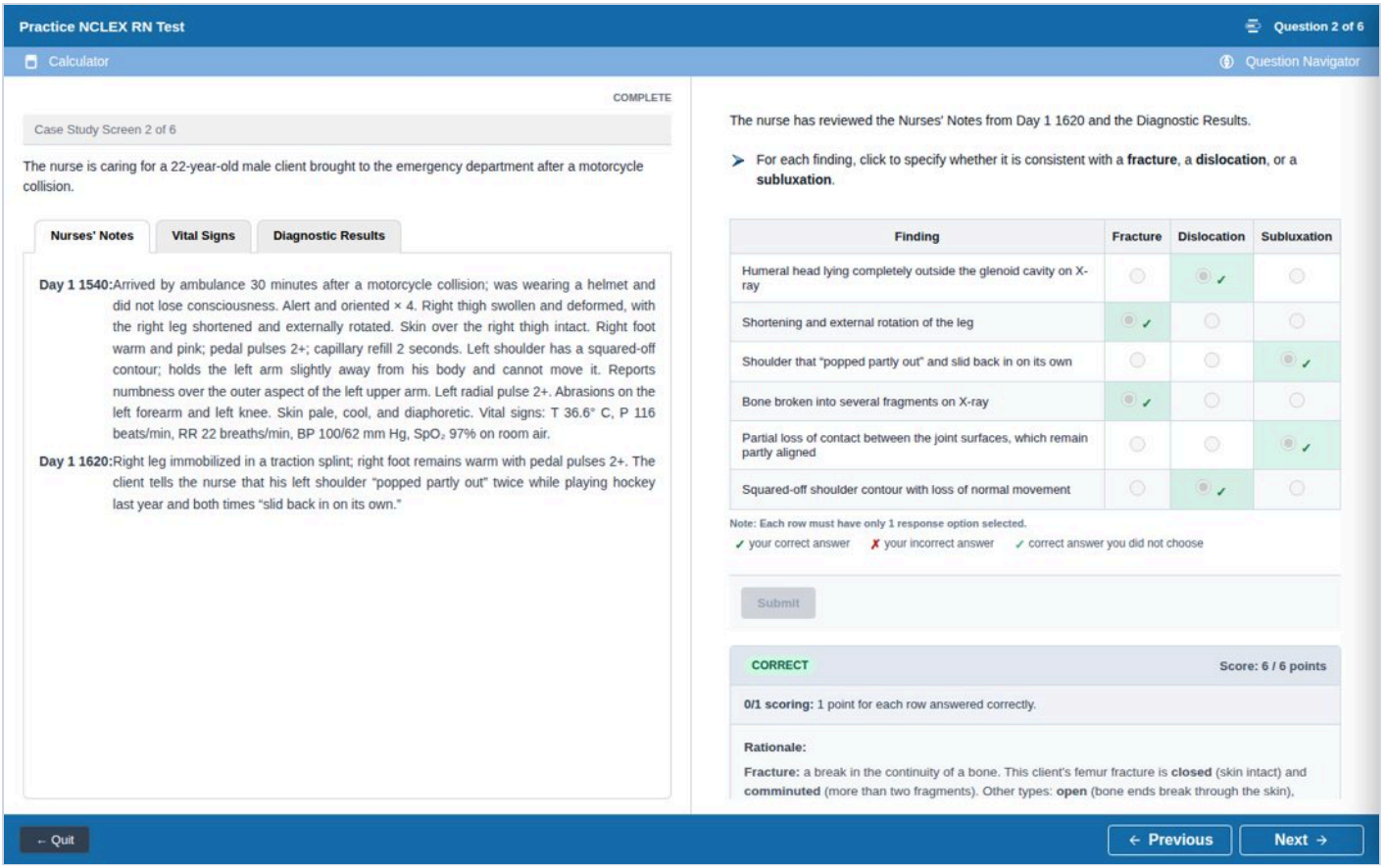
Tachycardia (116 beats/min), BP 100/62 mm Hg, and pale, cool, diaphoretic skin: a closed femoral shaft fracture can hide 1 to 1.5 L of blood in the thigh. These are early signs of hypovolemic shock and must be reported now.

Squared-off shoulder that he cannot move: the humeral head has left the joint (anterior dislocation is the most common type). A dislocation is reduced as soon as possible to protect the nerves, blood vessels and joint cartilage.

Numbness over the outer upper arm: the axillary nerve wraps around the neck of the humerus and is the nerve most often injured by an anterior shoulder dislocation. Sensation is documented before and after reduction.

Do not need immediate follow-up: intact skin over the fracture (a closed fracture), a warm, pink foot with strong pulses and brisk capillary refill, a strong left radial pulse, a temperature of 36.6° C, SpO_2 97% and an RR of 22 breaths/min (expected with pain). The abrasions need cleaning but are not urgent.

Case Study 2 (Fracture, dislocation, fat embolism and bone healing), screen 2 of 6: Analyze cues



Case Study 2, screen 2: rationale

CORRECT

Score: 6 / 6 points

0/1 scoring: 1 point for each row answered correctly.

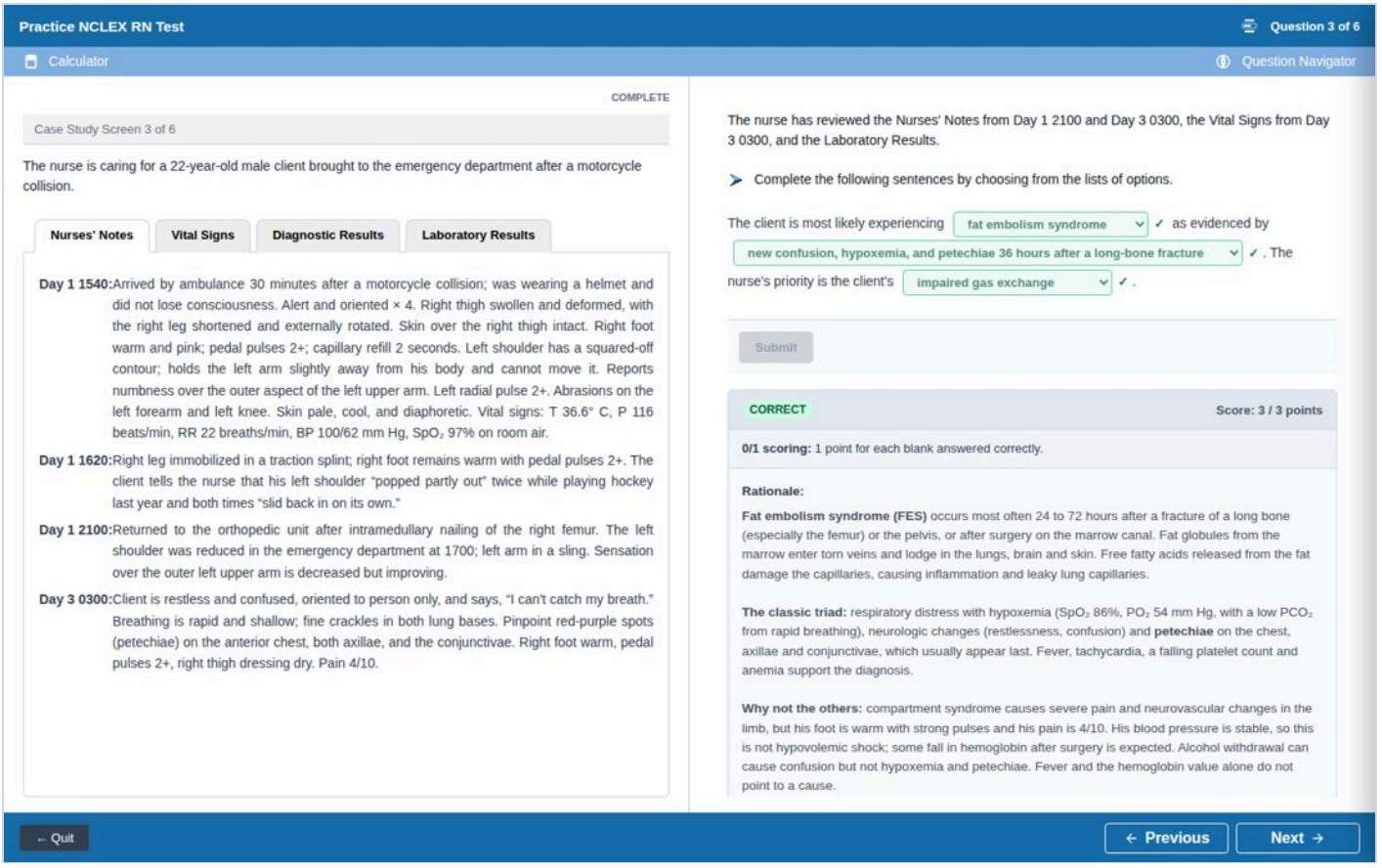
Rationale:

Fracture: a break in the continuity of a bone. This client's femur fracture is **closed** (skin intact) and **comminuted** (more than two fragments). Other types: **open** (bone ends break through the skin), **transverse** (straight across), **oblique** (at an angle), **spiral** (twisting around the shaft from a rotational force), **greenstick** (incomplete, on one side of a bendable child's bone), **compression** (bone crushed, as in vertebrae), **impacted** (one fragment driven into another), **avulsion** (a fragment pulled off by a tendon or ligament) and **pathologic** (through bone weakened by disease). Muscle spasm pulls the overriding femur fragments together, shortening and rotating the leg.

Dislocation: the joint surfaces lose contact **completely**, as the X-ray shows for the left shoulder. The joint looks deformed (the shoulder is squared off because the humeral head no longer fills out the deltoid) and cannot move normally.

Subluxation: a **partial** dislocation; the joint surfaces still touch in part. It often reduces on its own, as in his hockey injuries. Repeated subluxations stretch the capsule and ligaments, which made this full dislocation more likely.

Case Study 2 (Fracture, dislocation, fat embolism and bone healing), screen 3 of 6: Prioritize hypotheses



Case Study 2, screen 3: rationale

CORRECT **Score: 3 / 3 points**

0/1 scoring: 1 point for each blank answered correctly.

Rationale:

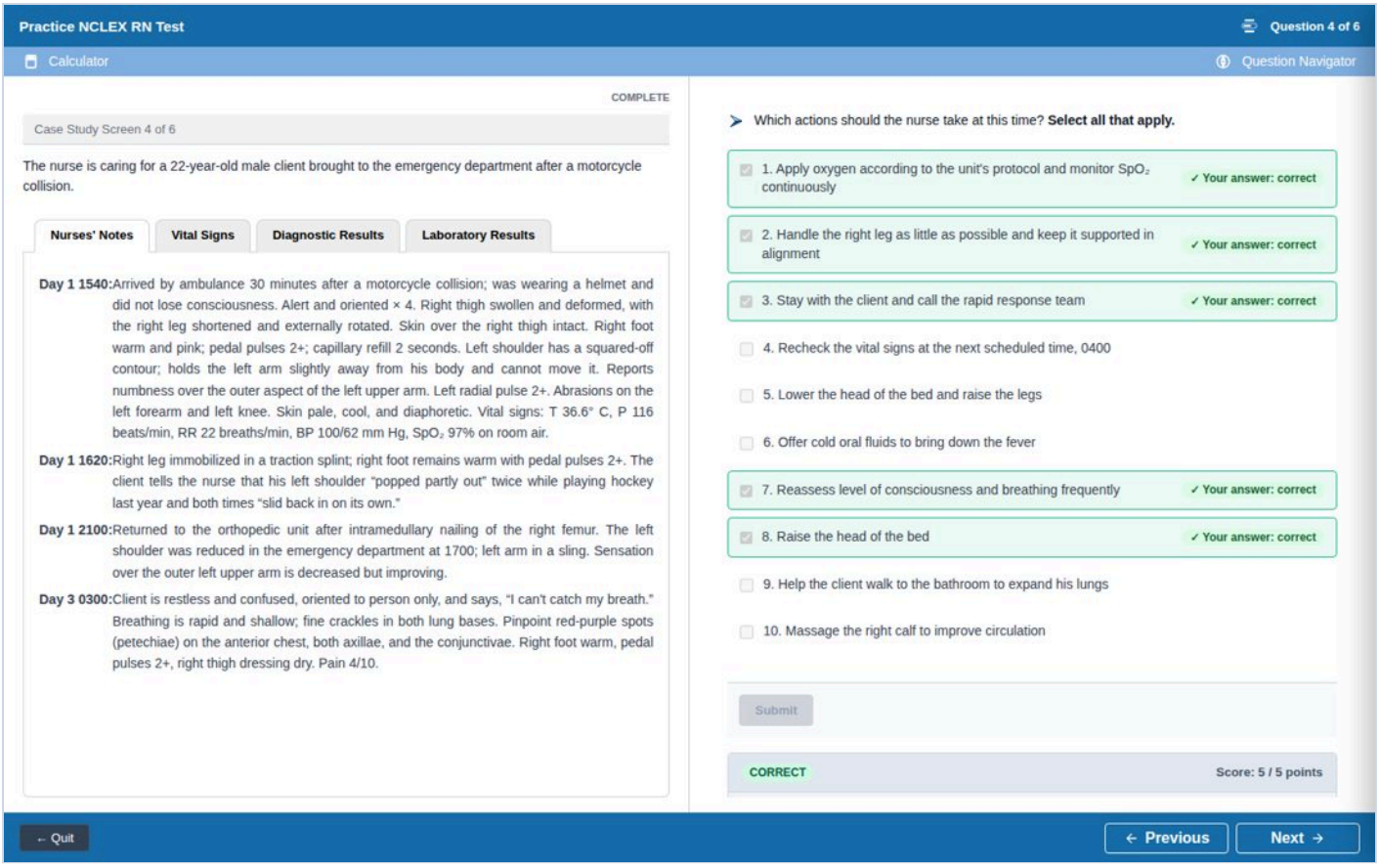
Fat embolism syndrome (FES) occurs most often 24 to 72 hours after a fracture of a long bone (especially the femur) or the pelvis, or after surgery on the marrow canal. Fat globules from the marrow enter torn veins and lodge in the lungs, brain and skin. Free fatty acids released from the fat damage the capillaries, causing inflammation and leaky lung capillaries.

The classic triad: respiratory distress with hypoxemia (SpO_2 86%, PO_2 54 mm Hg, with a low PCO_2 from rapid breathing), neurologic changes (restlessness, confusion) and **petechiae** on the chest, axillae and conjunctivae, which usually appear last. Fever, tachycardia, a falling platelet count and anemia support the diagnosis.

Why not the others: compartment syndrome causes severe pain and neurovascular changes in the limb, but his foot is warm with strong pulses and his pain is 4/10. His blood pressure is stable, so this is not hypovolemic shock; some fall in hemoglobin after surgery is expected. Alcohol withdrawal can cause confusion but not hypoxemia and petechiae. Fever and the hemoglobin value alone do not point to a cause.

Priority: airway and breathing come first. The client's oxygenation is failing (impaired gas exchange), which also explains his confusion. Pain, infection risk and mobility matter but are not life-threatening now.

Case Study 2 (Fracture, dislocation, fat embolism and bone healing), screen 4 of 6: Generate solutions



Case Study 2, screen 4: rationale

CORRECT

Score: 5 / 5 points

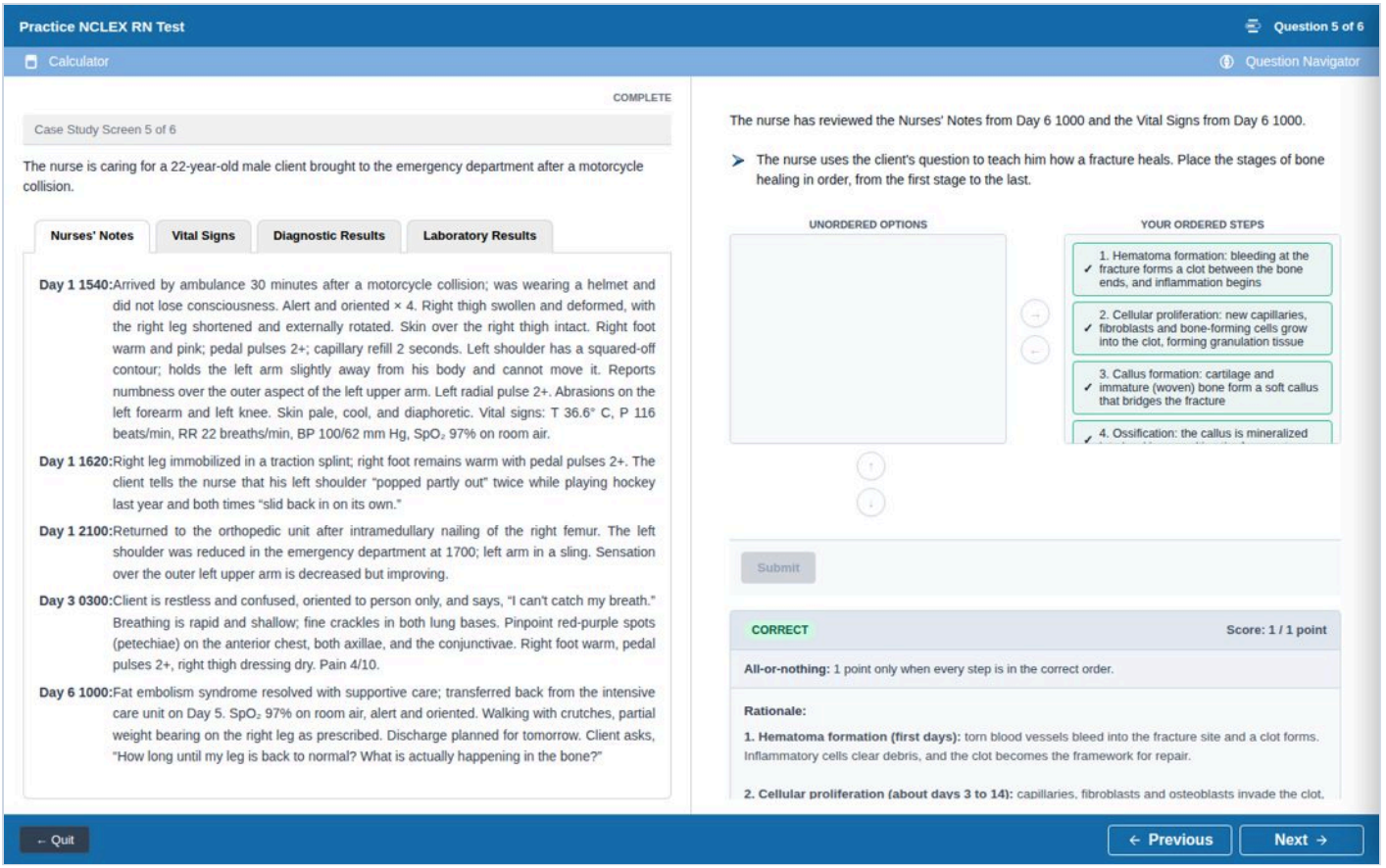
+/− scoring: each correct choice earns 1 point and each incorrect choice takes 1 point away (never below 0). You selected 5 correct (+5) and 0 incorrect (−0) = 5 points.

Rationale:

Correct: the client's oxygenation and level of consciousness are deteriorating, so the nurse stays with him and calls the rapid response team. Raising the head of the bed improves lung expansion and oxygenation. Oxygen is started under the unit's protocol, with continuous pulse oximetry. Frequent reassessment shows whether he is improving or needs airway support. Moving the fractured limb as little as possible limits further release of fat from the marrow.

Incorrect: massaging the calf could dislodge a clot if a deep vein thrombosis is present and does nothing for the lungs. Lowering the head of the bed worsens breathing. A confused, hypoxemic client with a newly repaired femur must not walk. Oral fluids risk aspiration in a confused, breathless client. Waiting for the next scheduled check delays care while he deteriorates.

Case Study 2 (Fracture, dislocation, fat embolism and bone healing), screen 5 of 6: Take action



Case Study 2, screen 5: rationale

CORRECT **Score: 1 / 1 point**

All-or-nothing: 1 point only when every step is in the correct order.

Rationale:

1. Hematoma formation (first days): torn blood vessels bleed into the fracture site and a clot forms. Inflammatory cells clear debris, and the clot becomes the framework for repair.

2. Cellular proliferation (about days 3 to 14): capillaries, fibroblasts and osteoblasts invade the clot, turning it into granulation tissue.

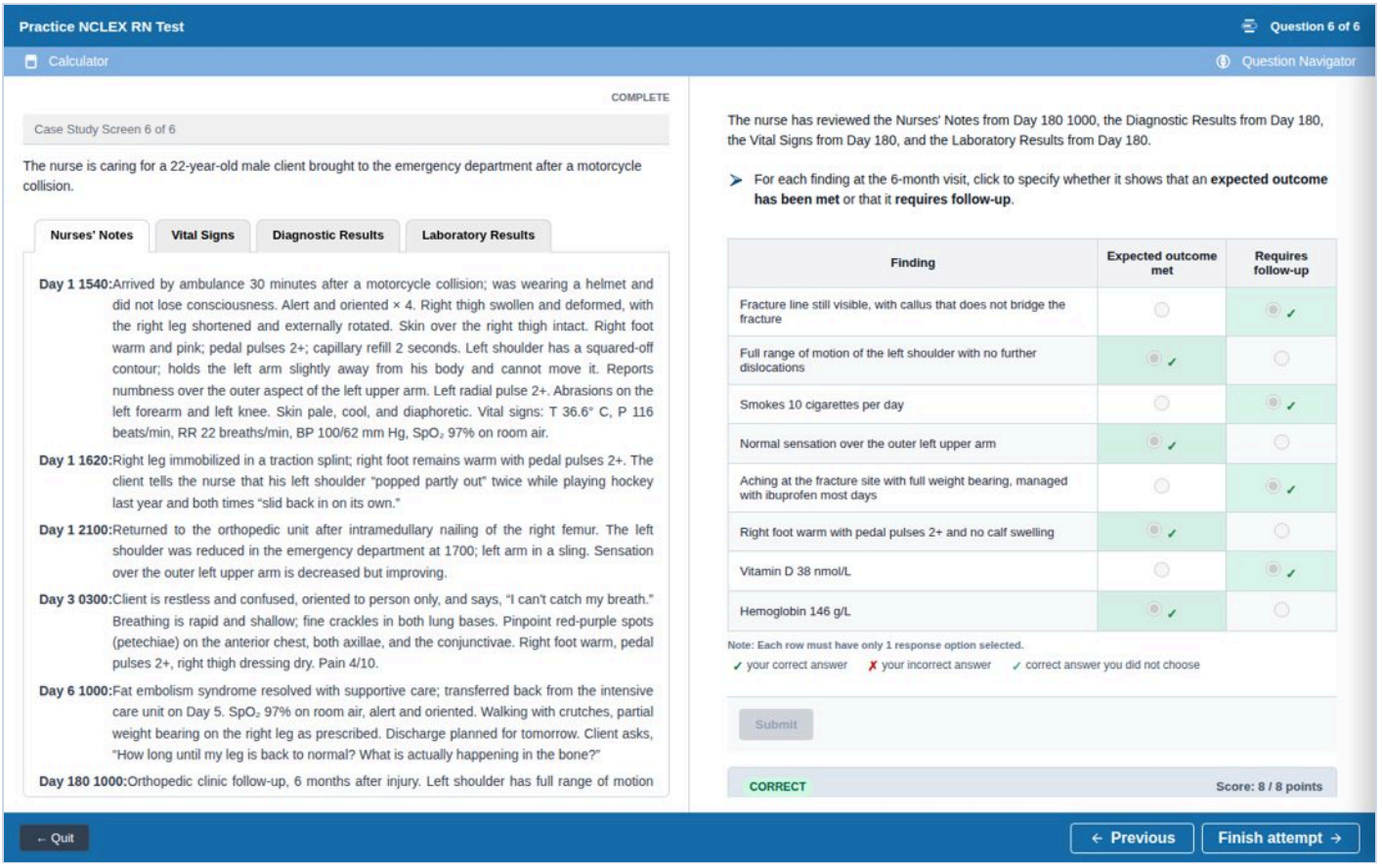
3. Callus formation (weeks 2 to 6): cartilage and woven bone form a callus around and between the bone ends. The fracture is held together but is not yet strong, which is why weight bearing is limited.

4. Ossification (weeks 3 to 12 or more): minerals are laid down and the callus becomes hard bone. On X-ray, bridging bone crosses the fracture line.

5. Remodeling (months to a year or more): osteoclasts remove the excess callus and the bone is reshaped along the lines of stress, restoring near-normal strength.

Healing times vary with the bone, the age of the client, the blood supply and other factors. A large weight-bearing bone such as the femur typically takes several months to unite. Smoking, poor nutrition, low vitamin D, infection, diabetes, corticosteroids and movement at the fracture site all slow healing.

Case Study 2 (Fracture, dislocation, fat embolism and bone healing), screen 6 of 6: Evaluate outcomes



Case Study 2, screen 6: rationale

CORRECT

Score: 8 / 8 points

0/1 scoring: 1 point for each row answered correctly.

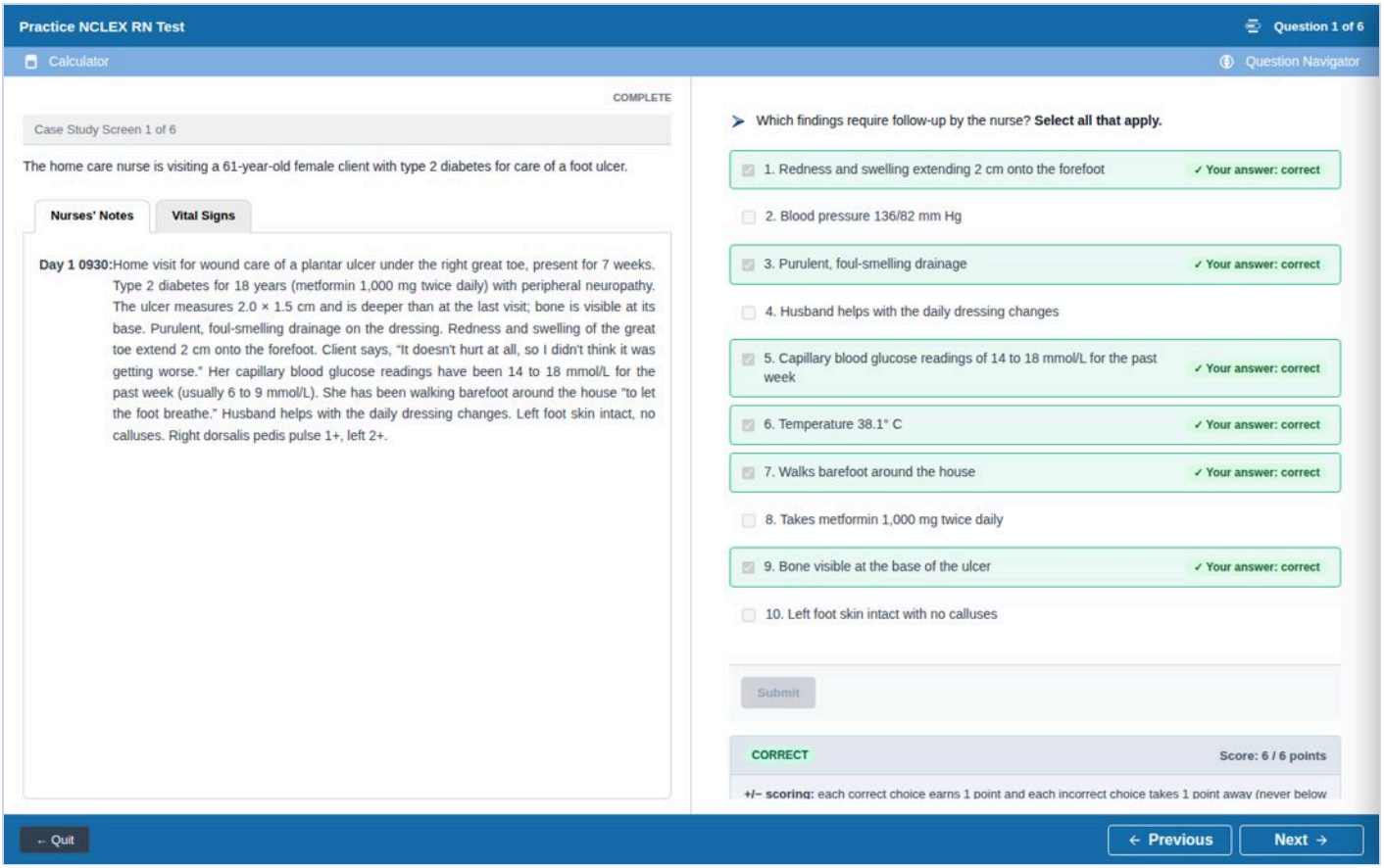
Rationale:

Expected outcomes met: the shoulder has healed without recurrence and the axillary nerve has recovered. Circulation to the right leg is normal with no signs of deep vein thrombosis. The hemoglobin has returned to normal after the blood loss.

Requires follow-up: delayed union. By 6 months a femur would be expected to show bridging bone. A fracture that heals more slowly than expected is a **delayed union**; if healing stops altogether it becomes a **nonunion**; a fracture that heals in a poor position is a **malunion**. Pain at the fracture site with weight bearing is a sign that the bone has not united.

Factors slowing his healing, which the nurse reports and addresses: smoking (nicotine constricts blood vessels and reduces oxygen delivery to the healing bone); **low vitamin D** (needed to absorb calcium for mineralizing the callus); and regular **NSAID** use (evidence suggests NSAIDs may interfere with the inflammatory phase of bone healing). The nurse reports these findings to the provider, offers smoking cessation support, and reinforces nutrition with enough protein, calcium and vitamin D.

Case Study 3 (Osteomyelitis), screen 1 of 6: Recognize cues



Case Study 3, screen 1: rationale

CORRECT **Score: 6 / 6 points**

+/− scoring: each correct choice earns 1 point and each incorrect choice takes 1 point away (never below 0). You selected 6 correct (+6) and 0 incorrect (−0) = 6 points.

Rationale:

Require follow-up:

Bone visible (or felt with a probe) at the base of a foot ulcer strongly suggests that the infection has reached the bone (osteomyelitis).

Purulent, foul-smelling drainage and spreading redness and swelling show active infection of the soft tissue around the ulcer. A swollen, red, “sausage-shaped” toe is typical when the bone underneath is infected.

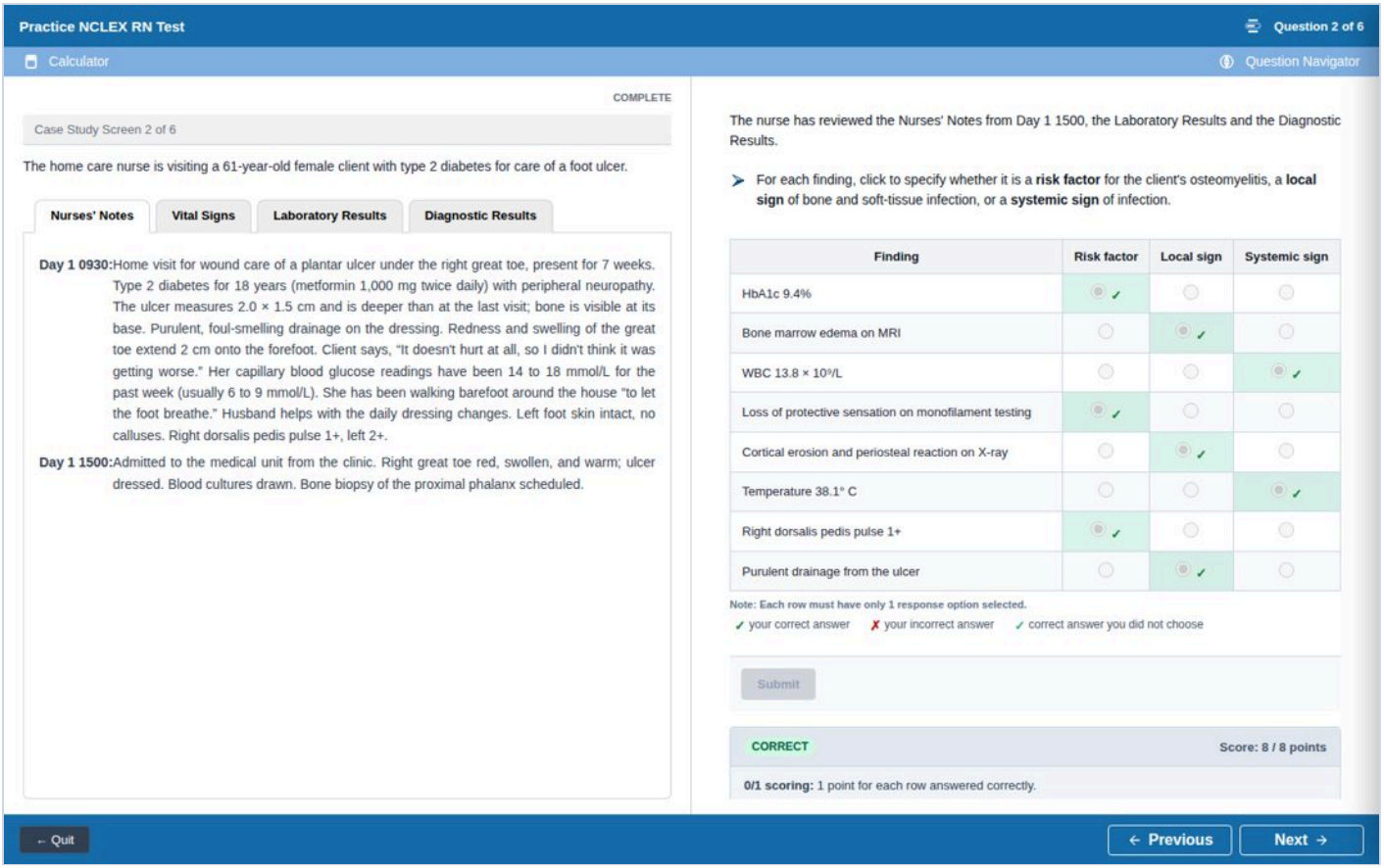
Fever shows a systemic response to the infection.

High blood glucose: infection raises stress hormones (cortisol, catecholamines), which raise blood glucose. Persistent hyperglycemia in turn impairs neutrophil function and wound healing, so the two worsen each other.

Walking barefoot on an insensate foot puts pressure and trauma on the ulcer. The client's lack of pain is not reassuring: peripheral neuropathy removes the warning sign, so infection can advance unnoticed.

Do not need follow-up: her blood pressure is acceptable, the husband's help is a strength, metformin is her usual medication, and the left foot is intact. The weaker right dorsalis pedis pulse is worth noting as a sign of reduced blood flow.

Case Study 3 (Osteomyelitis), screen 2 of 6: Analyze cues



Case Study 3, screen 2: rationale

CORRECT **Score: 8 / 8 points**

0/1 scoring: 1 point for each row answered correctly.

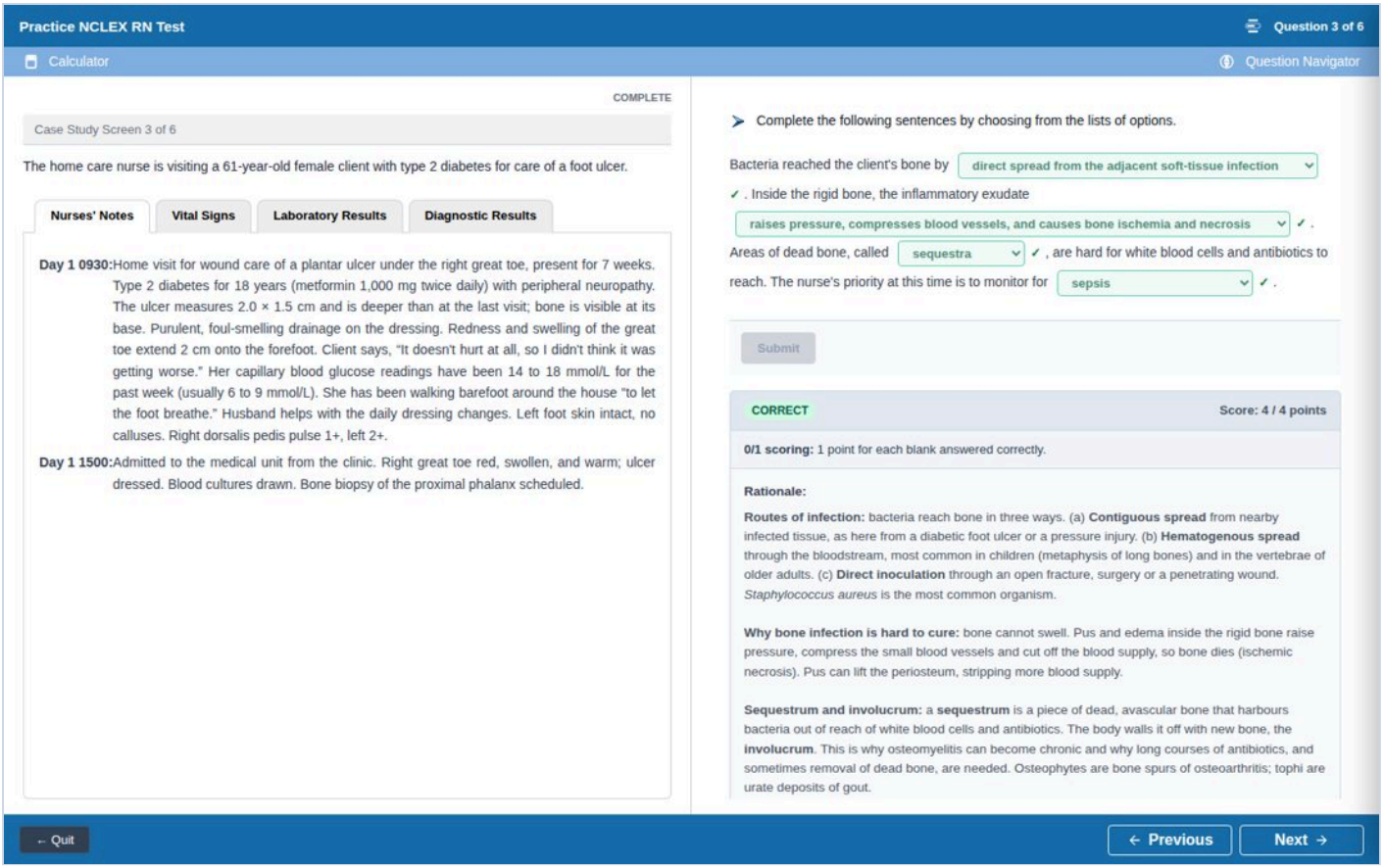
Rationale:

Risk factors: poorly controlled diabetes (HbA1c 9.4%) impairs white blood cell function and healing. Neuropathy (loss of protective sensation) lets ulcers form and worsen unnoticed. Reduced arterial blood flow (weak pedal pulse) delivers less oxygen, white blood cells and antibiotic to the tissue.

Local signs: purulent drainage shows infection at the site. The MRI shows marrow edema early, while plain X-rays show bone destruction (cortical erosion) and new bone laid down by the periosteum only after 10 to 14 days of infection.

Systemic signs: fever and a raised WBC count reflect the body's response to infection. ESR and CRP are also raised (ESR 92 mm/h, CRP 118 mg/L); a very high ESR in a diabetic foot ulcer suggests osteomyelitis, and both are followed to judge the response to treatment.

Case Study 3 (Osteomyelitis), screen 3 of 6: Prioritize hypotheses



Case Study 3, screen 3: rationale

CORRECT **Score: 4 / 4 points**

0/1 scoring: 1 point for each blank answered correctly.

Rationale:

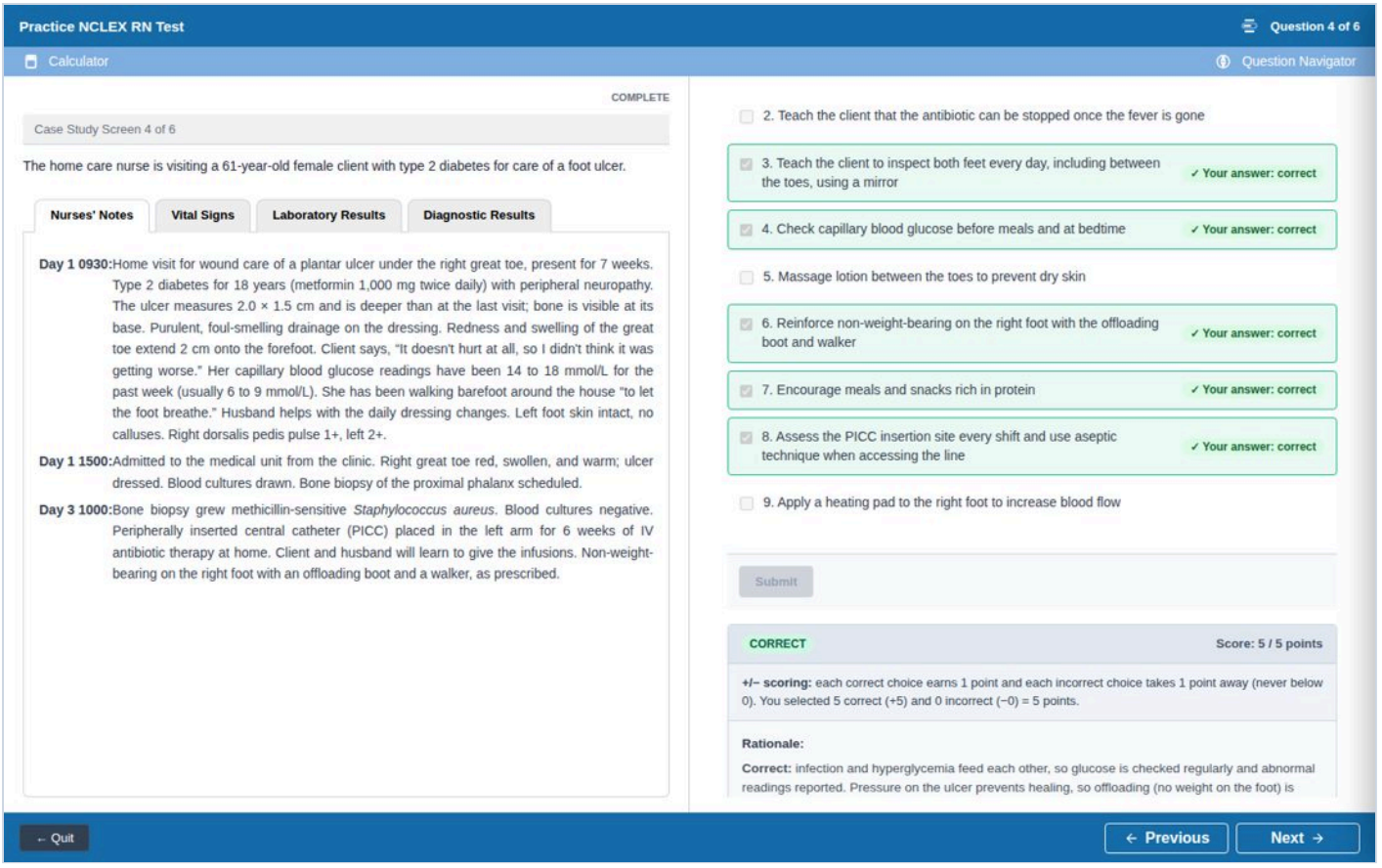
Routes of infection: bacteria reach bone in three ways. (a) **Contiguous spread** from nearby infected tissue, as here from a diabetic foot ulcer or a pressure injury. (b) **Hematogenous spread** through the bloodstream, most common in children (metaphysis of long bones) and in the vertebrae of older adults. (c) **Direct inoculation** through an open fracture, surgery or a penetrating wound. *Staphylococcus aureus* is the most common organism.

Why bone infection is hard to cure: bone cannot swell. Pus and edema inside the rigid bone raise pressure, compress the small blood vessels and cut off the blood supply, so bone dies (ischemic necrosis). Pus can lift the periosteum, stripping more blood supply.

Sequestrum and involucrum: a **sequestrum** is a piece of dead, avascular bone that harbours bacteria out of reach of white blood cells and antibiotics. The body walls it off with new bone, the **involucrum**. This is why osteomyelitis can become chronic and why long courses of antibiotics, and sometimes removal of dead bone, are needed. Osteophytes are bone spurs of osteoarthritis; tophi are urate deposits of gout.

Priority: the client has fever, a raised WBC count, and blood cultures pending. Spread of the infection into the bloodstream (**sepsis**) is the most immediate threat to her life. Fat embolism and compartment syndrome are complications of fractures, and osteoporosis is not an acute problem.

Case Study 3 (Osteomyelitis), screen 4 of 6: Generate solutions



Case Study 3, screen 4: rationale

CORRECT

Score: 5 / 5 points

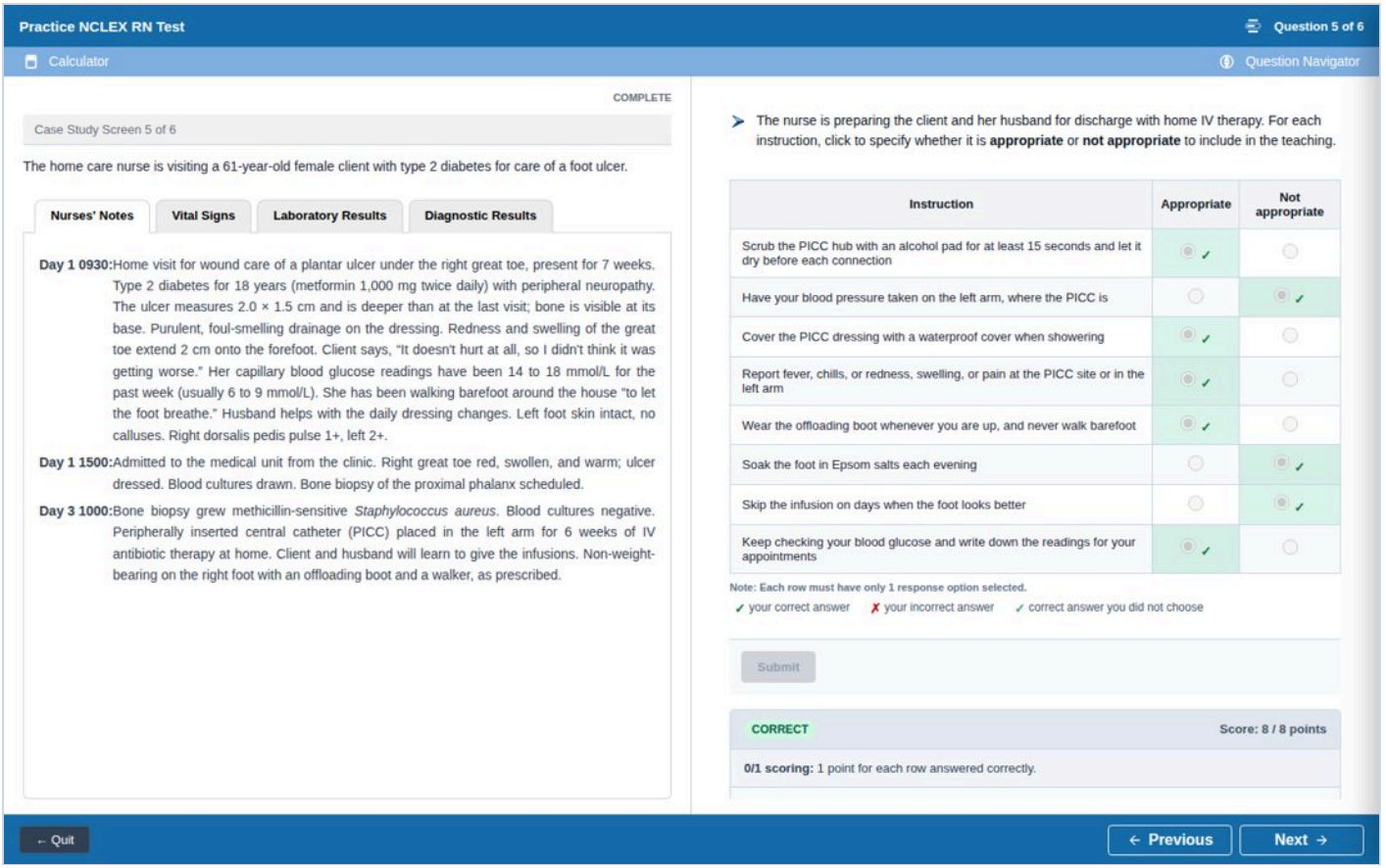
+/− scoring: each correct choice earns 1 point and each incorrect choice takes 1 point away (never below 0). You selected 5 correct (+5) and 0 incorrect (−0) = 5 points.

Rationale:

Correct: infection and hyperglycemia feed each other, so glucose is checked regularly and abnormal readings reported. Pressure on the ulcer prevents healing, so offloading (no weight on the foot) is essential. A PICC is a central line with a risk of bloodstream infection and thrombosis, so the site is assessed and accessed aseptically. Daily foot inspection is vital when sensation is lost, because the client cannot feel new injuries. Protein supports tissue repair and the immune response.

Incorrect: soaking macerates the skin and the wound and increases the risk of infection. Heating pads can burn a foot that cannot feel heat. Antibiotic courses for osteomyelitis are long (often 6 weeks) because bone is poorly perfused; stopping when the fever goes leaves bacteria in the bone and leads to chronic infection. Moisture between the toes promotes fungal infection and skin breakdown; lotion is applied to the tops and soles, not between the toes.

Case Study 3 (Osteomyelitis), screen 5 of 6: Take action



Case Study 3, screen 5: rationale

CORRECT

Score: 8 / 8 points

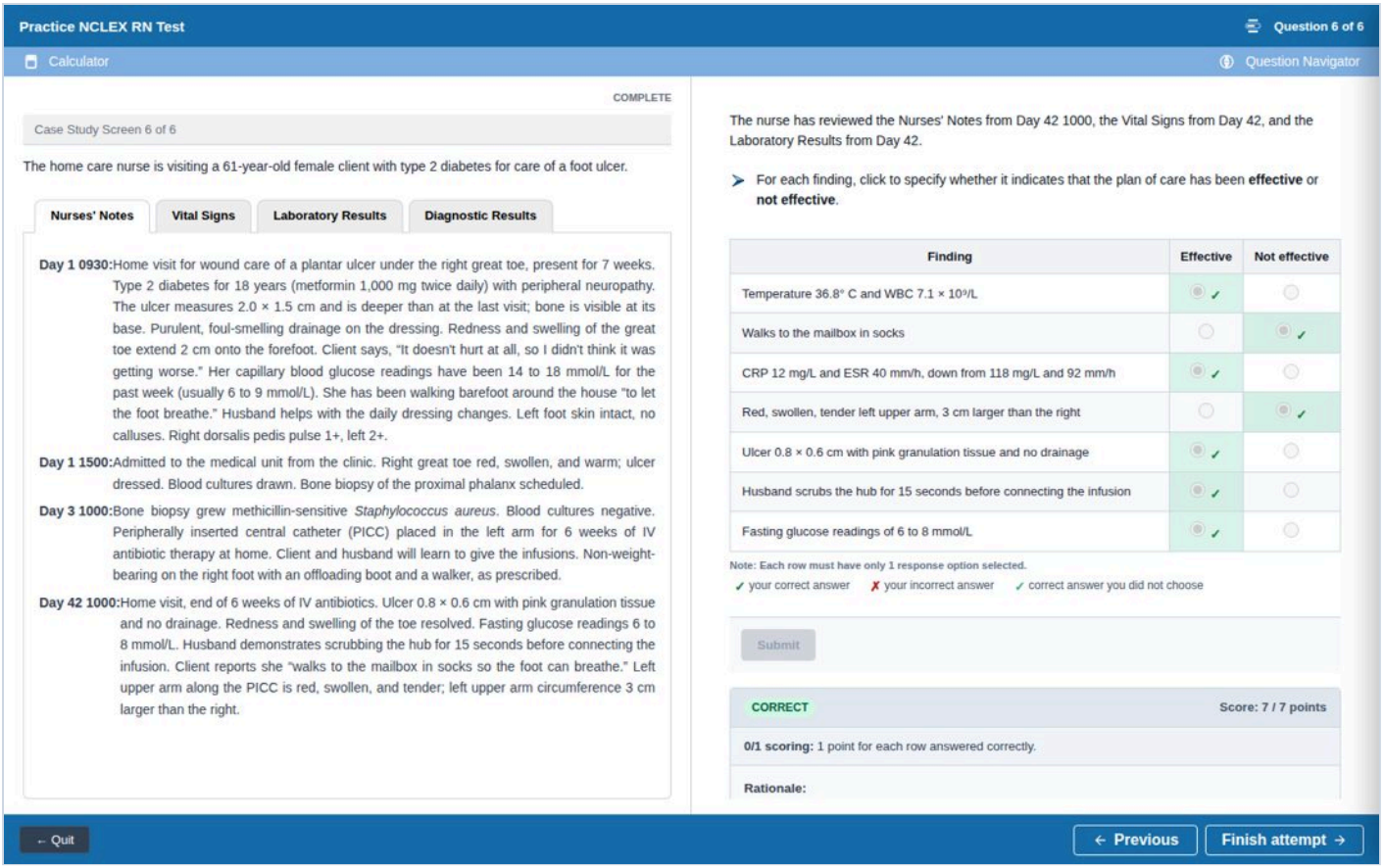
0/1 scoring: 1 point for each row answered correctly.

Rationale:

Appropriate: scrubbing the hub (“scrub the hub”) removes bacteria before each access and prevents central line infection. Keeping the dressing dry protects the site. Fever, chills, or redness, swelling or pain at the site or in the arm can mean line infection or a clot and are reported at once. Offloading and protective footwear protect the healing foot. Glucose records help the team adjust diabetes management, because good control supports healing.

Not appropriate: blood pressure cuffs, tourniquets and blood draws are avoided on the arm with the PICC (they can damage the catheter or the vein). Soaking macerates the wound. Missing doses lowers the drug level in bone and allows the infection to persist or become chronic.

Case Study 3 (Osteomyelitis), screen 6 of 6: Evaluate outcomes



Case Study 3, screen 6: rationale

CORRECT

Score: 7 / 7 points

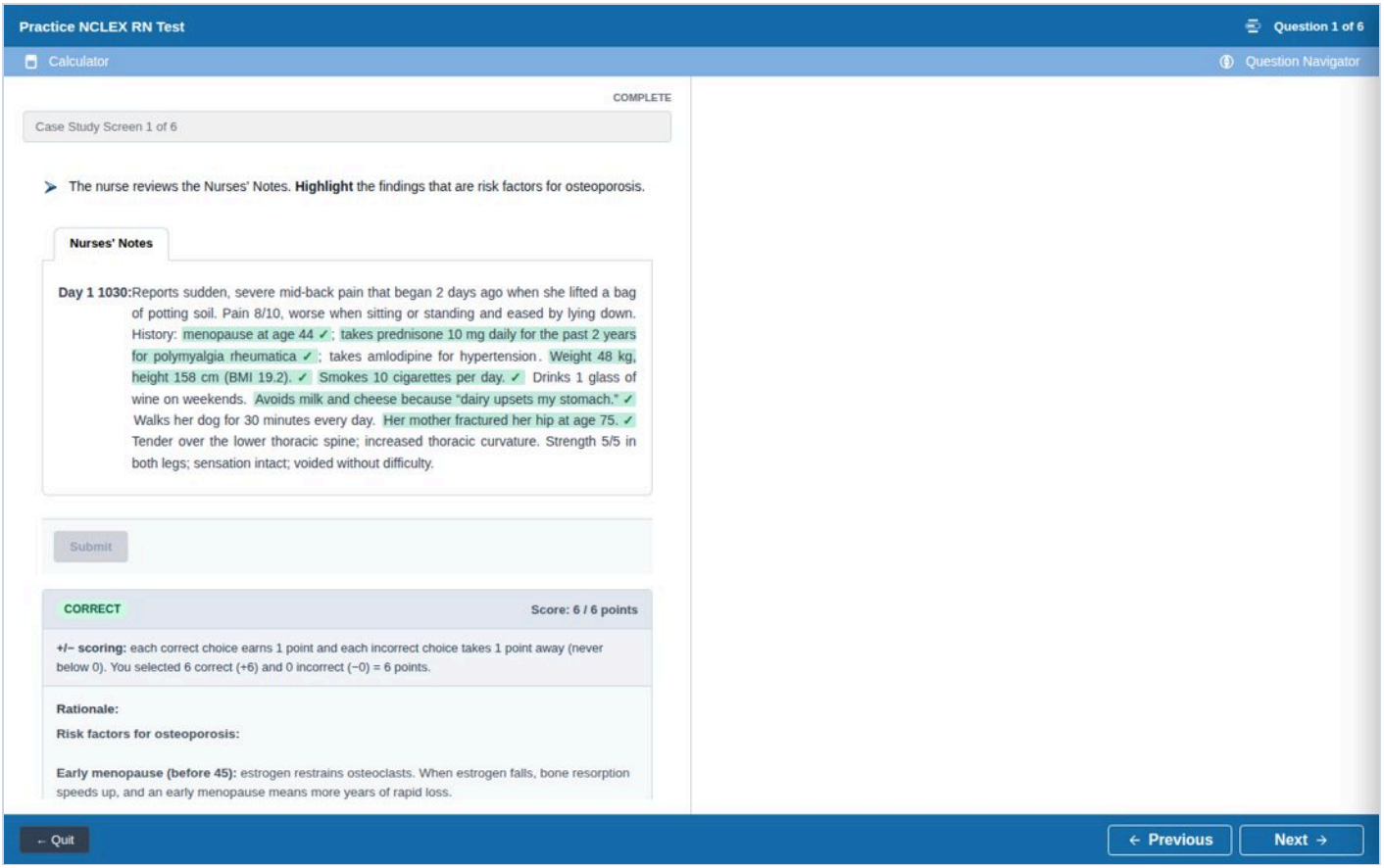
0/1 scoring: 1 point for each row answered correctly.

Rationale:

Effective: no fever and a normal WBC count; CRP and ESR falling steadily (ESR falls more slowly than CRP); a smaller, clean ulcer with healthy granulation tissue; improved glucose control; and correct line care by the husband.

Not effective: walking in socks puts pressure on the ulcer and exposes the insensate foot to injury; the nurse reinforces wearing the offloading boot. A red, swollen, tender arm with a larger circumference on the PICC side suggests a catheter-related deep vein thrombosis or phlebitis; the nurse does not use the line until it is assessed and reports the finding to the provider promptly.

Case Study 4 (Osteoporosis and vertebral compression fracture), screen 1 of 6: Recognize cues



Case Study 4, screen 1: rationale

CORRECT **Score: 6 / 6 points**

+/− scoring: each correct choice earns 1 point and each incorrect choice takes 1 point away (never below 0). You selected 6 correct (+6) and 0 incorrect (−0) = 6 points.

Rationale:

Risk factors for osteoporosis:

Early menopause (before 45): estrogen restrains osteoclasts. When estrogen falls, bone resorption speeds up, and an early menopause means more years of rapid loss.

Long-term glucocorticoids (7.5 mg/day or more of prednisone for 3 months or longer is the usual threshold): they suppress osteoblasts, increase resorption and reduce calcium absorption. This is the most common cause of secondary osteoporosis.

Low body weight (small frame, BMI under about 20): less mechanical load on the bones and less estrogen produced by fat tissue.

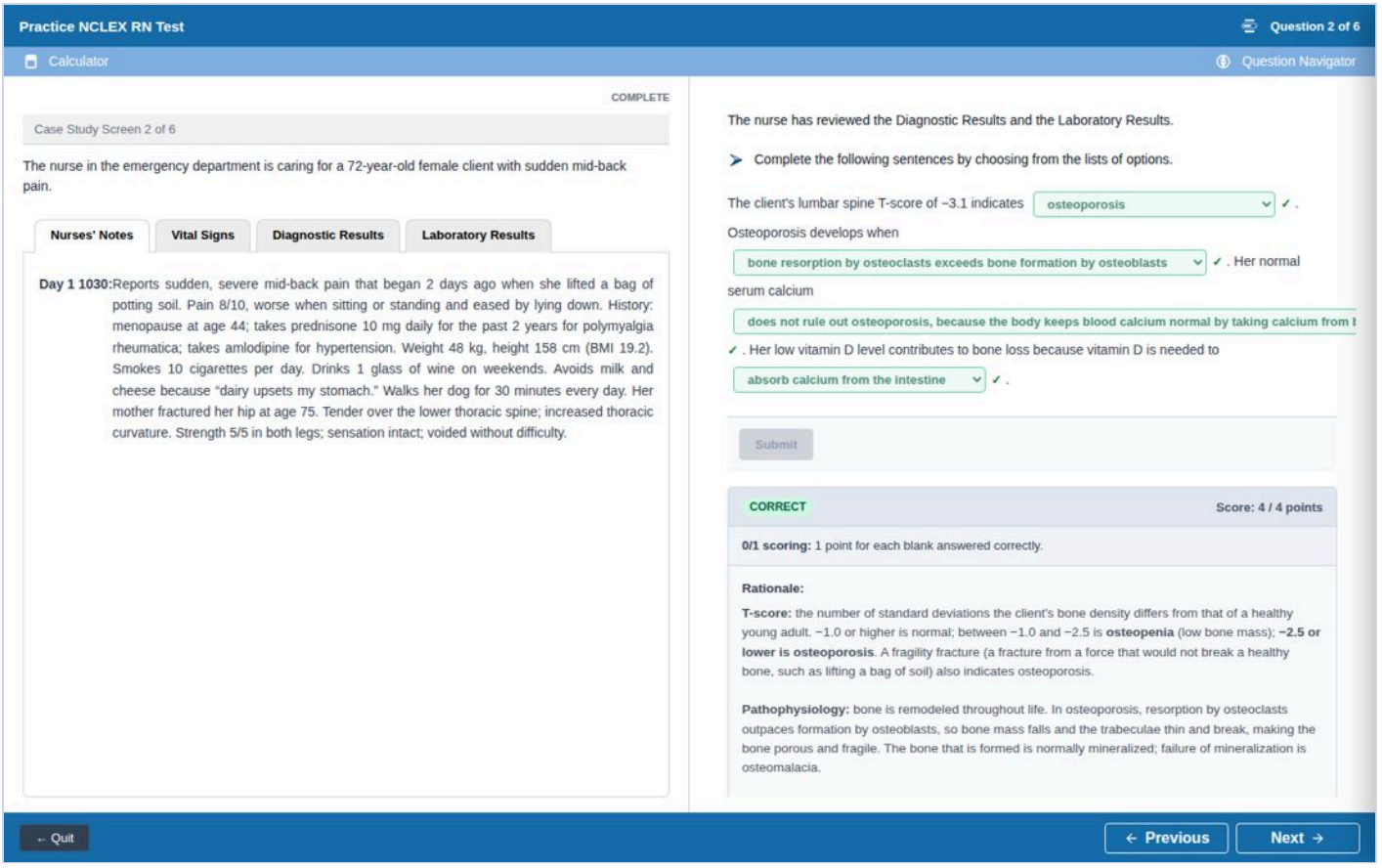
Smoking: toxic to osteoblasts and lowers estrogen levels.

Low calcium intake: avoiding dairy without other calcium sources means the body takes calcium from bone to keep blood levels normal.

Parental hip fracture: a strong family risk factor, reflecting inherited peak bone mass.

Not risk factors: one glass of wine on weekends (risk rises with 3 or more drinks a day), daily walking (weight-bearing exercise protects bone) and amlodipine.

Case Study 4 (Osteoporosis and vertebral compression fracture), screen 2 of 6: Analyze cues



Case Study 4, screen 2: rationale

CORRECT — Score: 4 / 4 points

0/1 scoring: 1 point for each blank answered correctly.

Rationale:

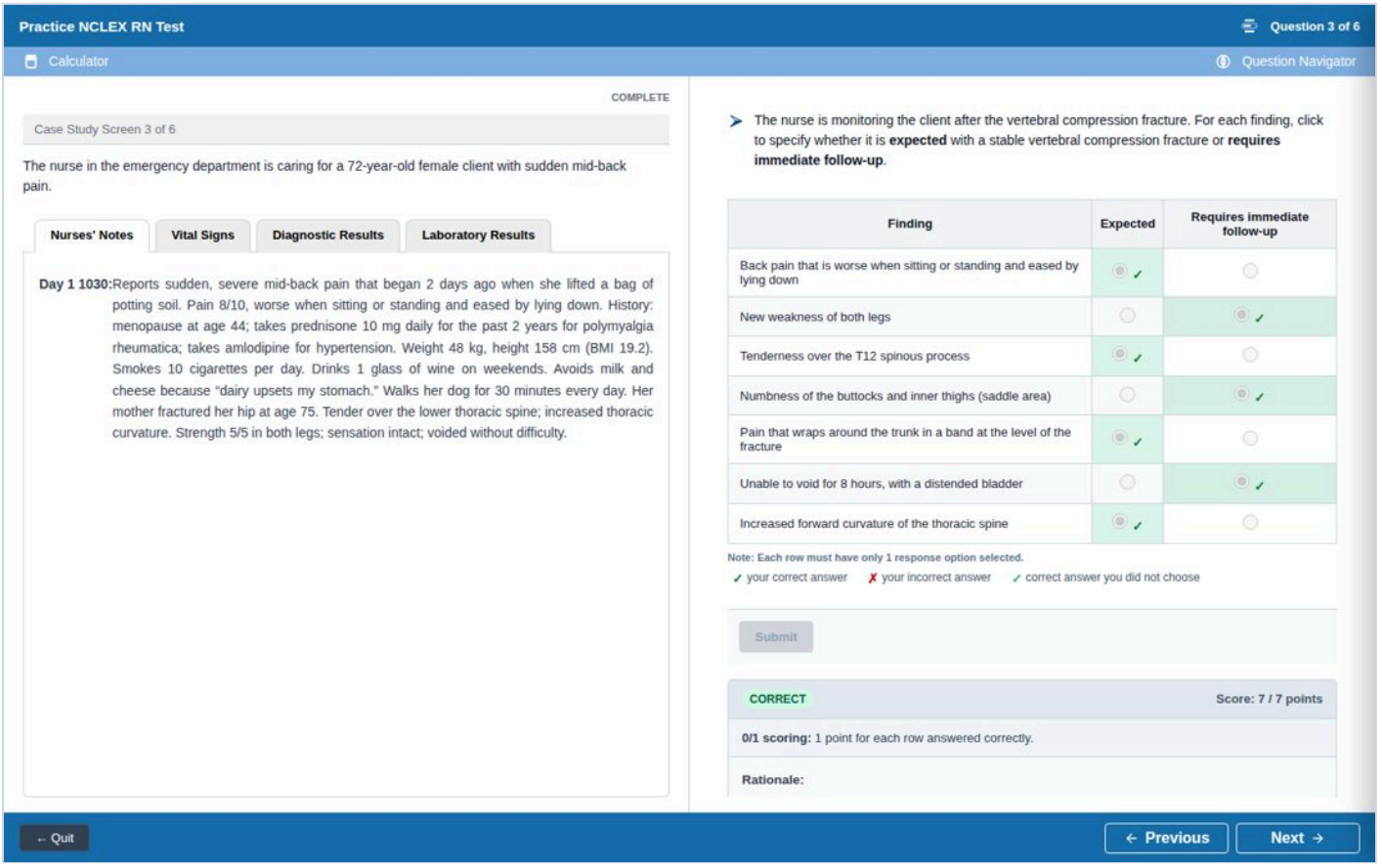
T-score: the number of standard deviations the client's bone density differs from that of a healthy young adult. −1.0 or higher is normal; between −1.0 and −2.5 is **osteopenia** (low bone mass); **−2.5 or lower is osteoporosis**. A fragility fracture (a fracture from a force that would not break a healthy bone, such as lifting a bag of soil) also indicates osteoporosis.

Pathophysiology: bone is remodeled throughout life. In osteoporosis, resorption by osteoclasts outpaces formation by osteoblasts, so bone mass falls and the trabeculae thin and break, making the bone porous and fragile. The bone that is formed is normally mineralized; failure of mineralization is osteomalacia.

Serum calcium is tightly controlled by parathyroid hormone, which releases calcium from bone when intake is low. A normal blood calcium therefore says nothing about bone density.

Vitamin D increases the absorption of calcium (and phosphate) from the intestine. When it is low, less calcium is absorbed, parathyroid hormone rises and more calcium is taken from bone.

Case Study 4 (Osteoporosis and vertebral compression fracture), screen 3 of 6: Prioritize hypotheses



Case Study 4, screen 3: rationale

CORRECT

Score: 7 / 7 points

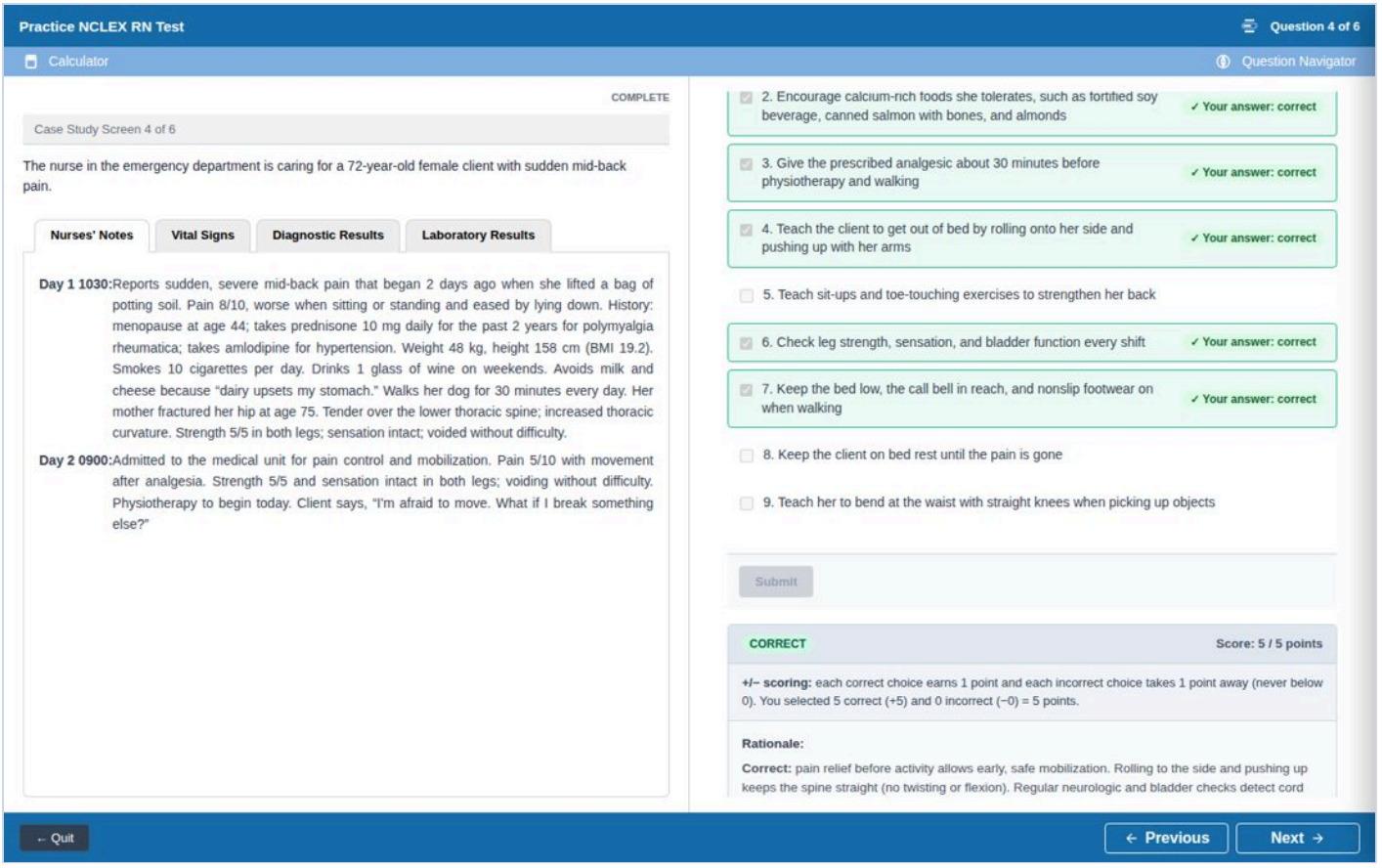
0/1 scoring: 1 point for each row answered correctly.

Rationale:

Expected: a compression fracture causes sudden, localized back pain over the fractured vertebra that worsens with upright posture (load on the spine) and eases when lying down. Irritation of the nerve roots at that level can cause pain that wraps around the trunk in a band. Repeated wedge fractures of the front of the thoracic vertebrae increase the forward curve (kyphosis, the “dowager's hump”) and cause loss of height.

Require immediate follow-up: new leg weakness, numbness in the saddle area, and urinary retention (or new incontinence) suggest that bone fragments or collapse are compressing the spinal cord or the cauda equina. These are emergencies; the nurse keeps the client flat, reports at once, and continues neurologic checks.

Case Study 4 (Osteoporosis and vertebral compression fracture), screen 4 of 6: Generate solutions



Case Study 4, screen 4: rationale

CORRECT

Score: 5 / 5 points

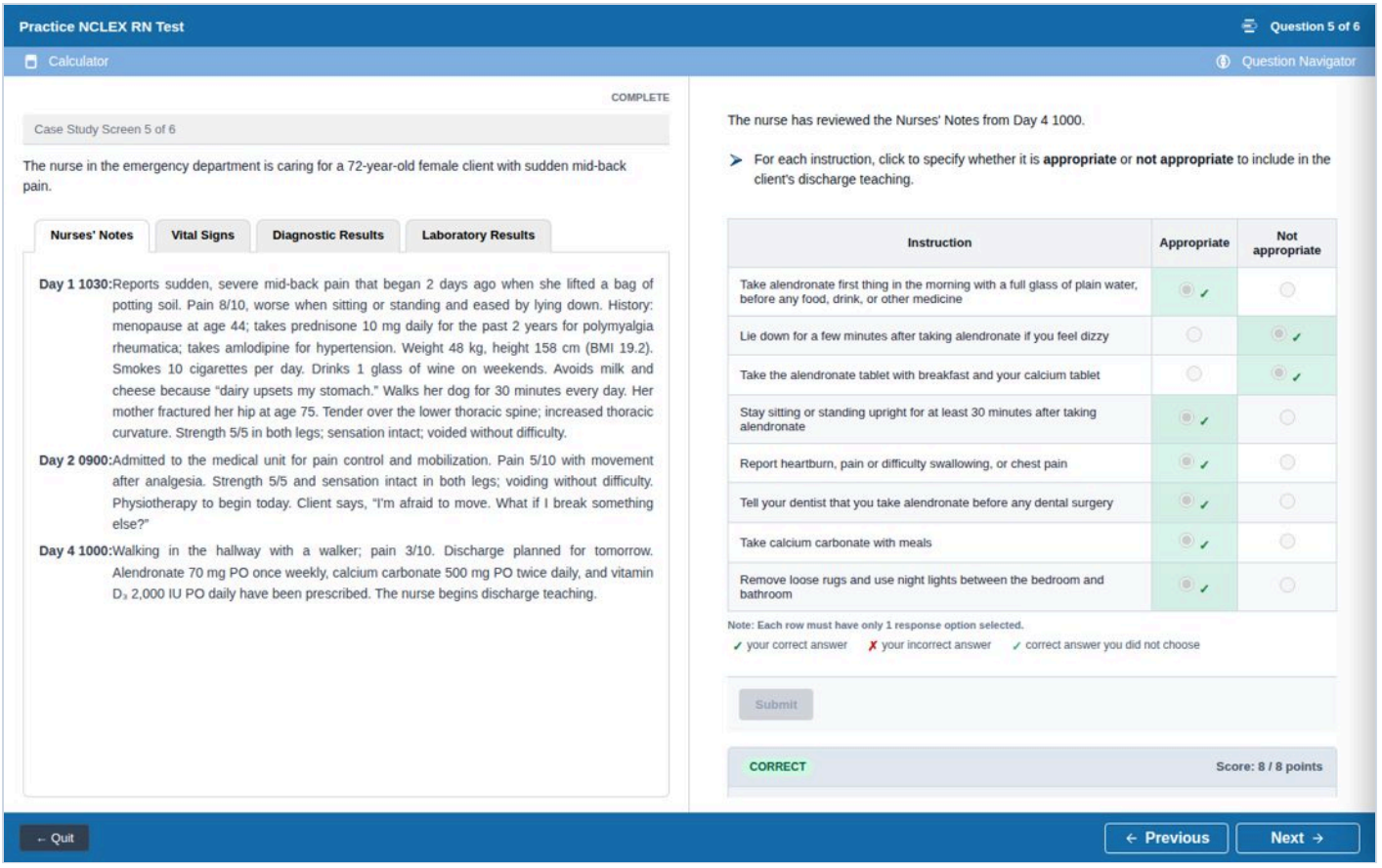
+/− scoring: each correct choice earns 1 point and each incorrect choice takes 1 point away (never below 0). You selected 5 correct (+5) and 0 incorrect (−0) = 5 points.

Rationale:

Correct: pain relief before activity allows early, safe mobilization. Rolling to the side and pushing up keeps the spine straight (no twisting or flexion). Regular neurologic and bladder checks detect cord compression early. Falls are the main cause of new fractures, so fall precautions are essential. Calcium intake matters, and lactose-free or non-dairy sources suit a client who avoids milk.

Incorrect: bed rest speeds up bone loss and causes deconditioning, pneumonia and blood clots; walking is encouraged as soon as pain allows. Bending at the waist and flexion exercises (sit-ups, toe touches) load the front of the vertebrae and can cause new compression fractures; lifting is done with bent knees and a straight back. A deep, soft chair forces the spine into flexion and makes standing up hard; a firm chair with arms is safer.

Case Study 4 (Osteoporosis and vertebral compression fracture), screen 5 of 6: Take action



Case Study 4, screen 5: rationale

CORRECT **Score: 8 / 8 points**

0/1 scoring: 1 point for each row answered correctly.

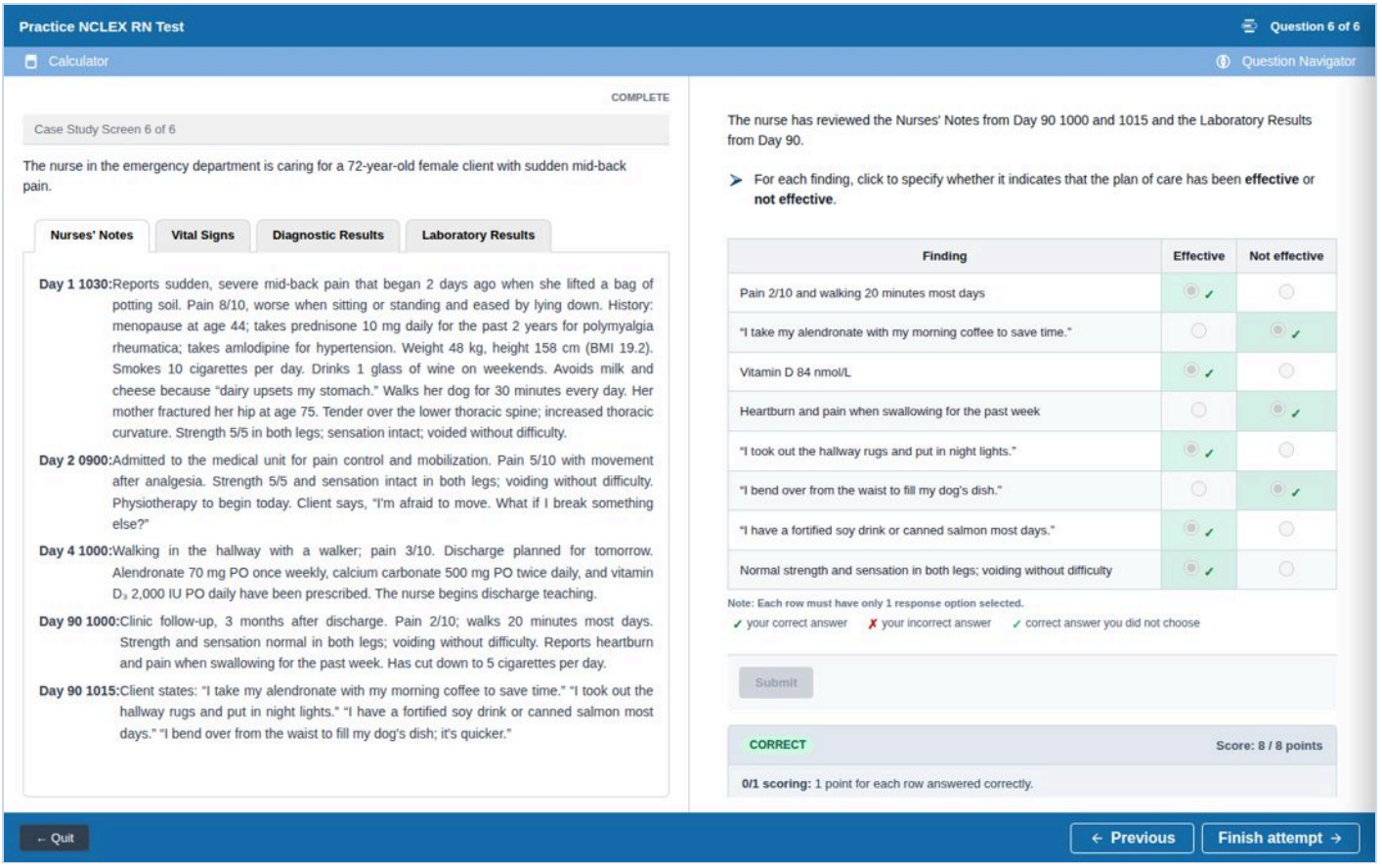
Rationale:

Alendronate (a bisphosphonate) slows bone resorption by osteoclasts. It is poorly absorbed and irritates the esophagus, so it is taken on an empty stomach with a full glass of plain water, at least 30 minutes before food, drinks (including coffee) or other medicines, and the client stays upright for at least 30 minutes. Lying down lets the tablet reflux and cause esophagitis or ulcers; taking it with food or calcium blocks its absorption. Heartburn, painful swallowing or chest pain can signal esophageal injury. Osteonecrosis of the jaw is a rare risk, so dentists should know before invasive dental work.

Calcium carbonate needs stomach acid to be absorbed, so it is taken with meals, and at a different time from alendronate.

Fall prevention at home (no loose rugs, night lights, grab bars, good lighting) prevents the falls that cause most osteoporotic fractures.

Case Study 4 (Osteoporosis and vertebral compression fracture), screen 6 of 6: Evaluate outcomes



Case Study 4, screen 6: rationale

CORRECT

Score: 8 / 8 points

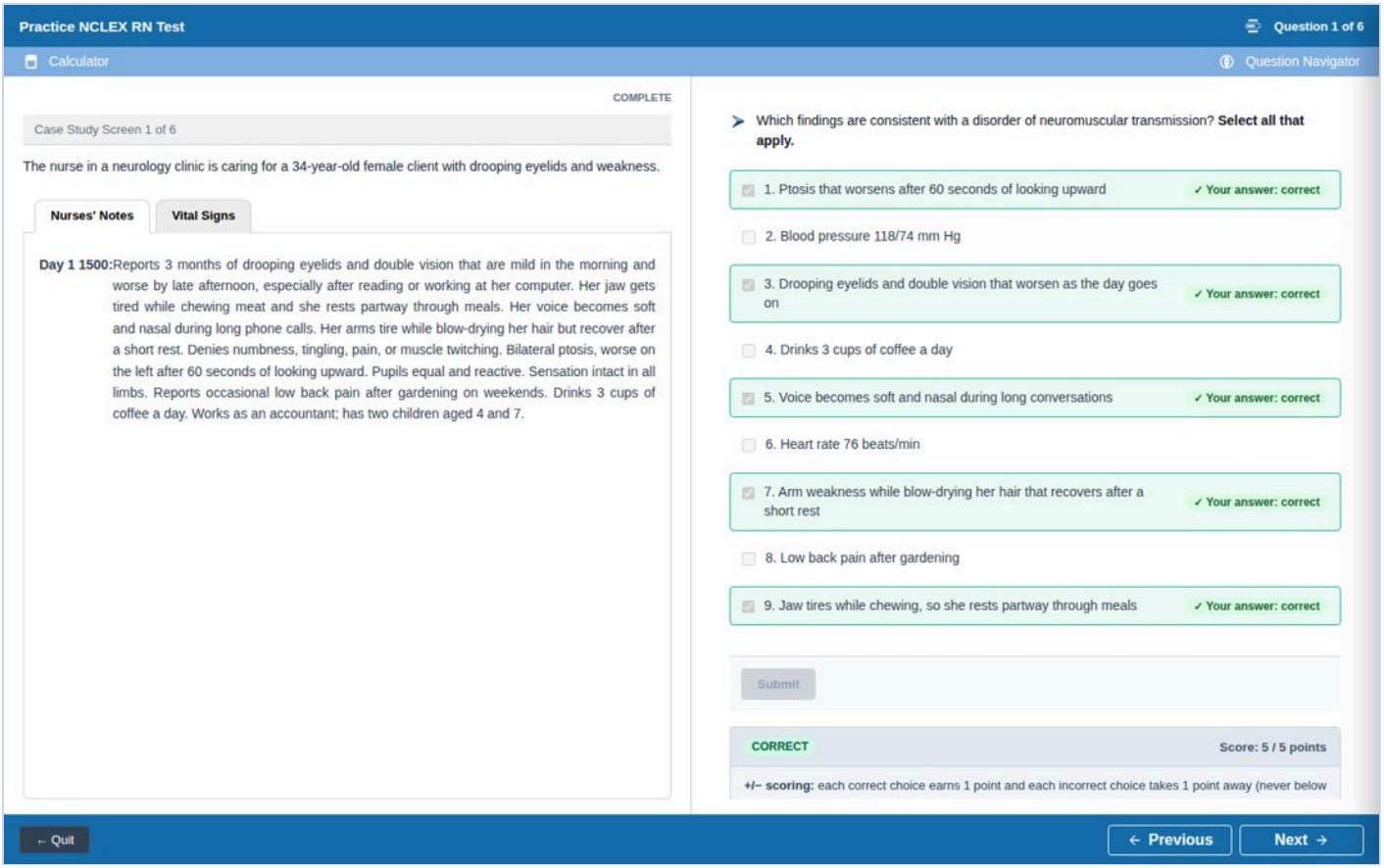
0/1 scoring: 1 point for each row answered correctly.

Rationale:

Effective: pain is controlled and she is walking regularly (weight-bearing exercise helps maintain bone). Her vitamin D is now in the normal range. She has made her home safer and includes calcium-rich foods she tolerates. No neurologic signs of cord compression.

Not effective: taking alendronate with coffee blocks its absorption and suggests she may not stay upright or wait before eating; together with her new heartburn and painful swallowing this points to esophageal irritation, which the nurse reports to the provider. Bending from the waist loads the front of the vertebrae and risks another compression fracture; the nurse reteaches bending at the knees with a straight back. Her reduced smoking is progress, and continued cessation support is offered.

Case Study 5 (Myasthenia gravis), screen 1 of 6: Recognize cues



Case Study 5, screen 1: rationale

CORRECT

Score: 5 / 5 points

+/− scoring: each correct choice earns 1 point and each incorrect choice takes 1 point away (never below 0). You selected 5 correct (+5) and 0 incorrect (−0) = 5 points.

Rationale:

Hallmark: fatigable weakness. In myasthenia gravis (MG), voluntary muscles weaken with repeated use and recover with rest. Weakness is therefore least in the morning and worst later in the day or after sustained activity.

Ocular muscles are affected first in most clients (ptosis, diplopia). Sustained upward gaze tires the eyelid muscle and brings out the ptosis.

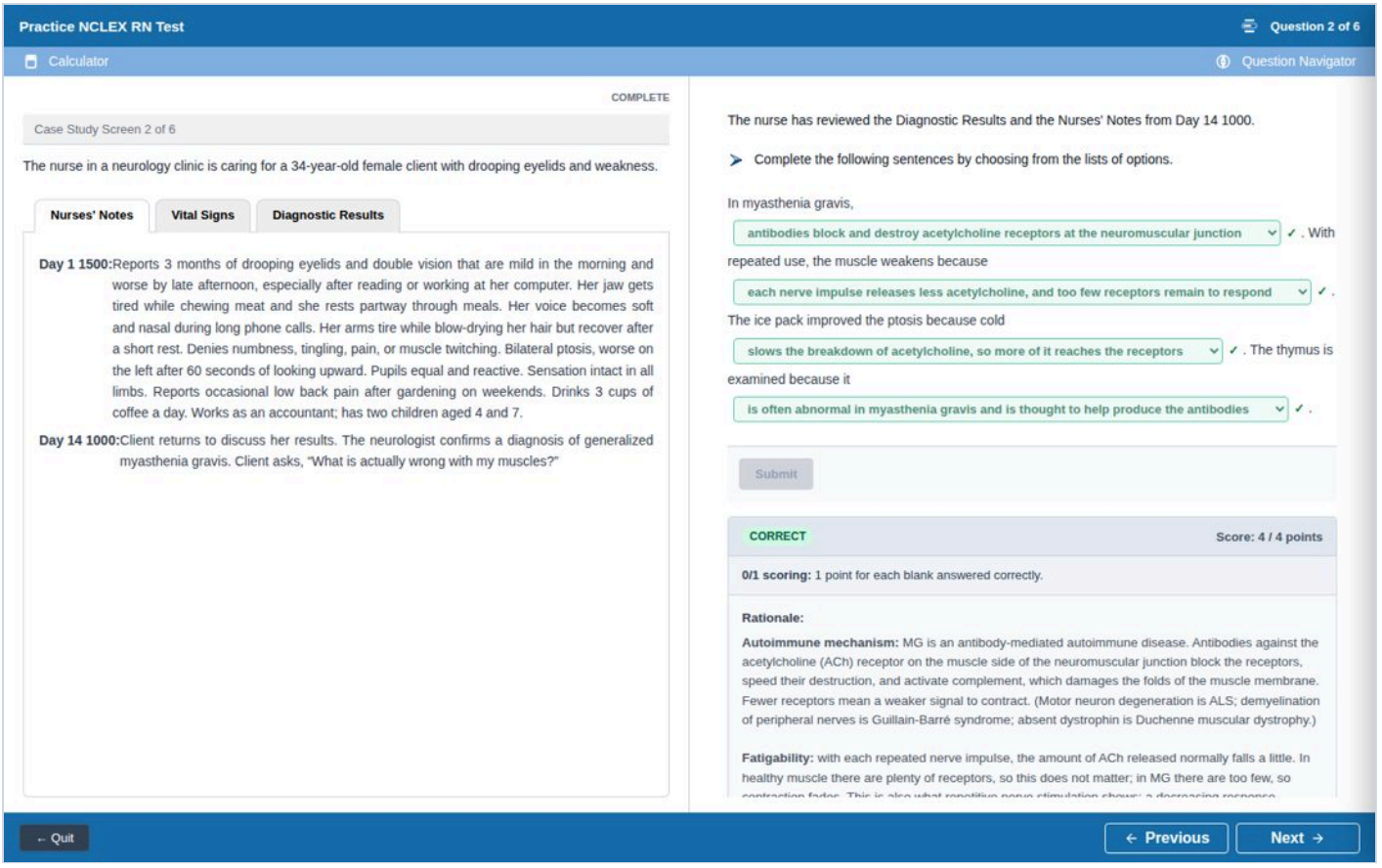
Bulbar muscles (chewing, swallowing, speaking) are often affected: a tiring jaw, a soft nasal voice, choking. Bulbar weakness matters because it threatens the airway.

Proximal limb muscles tire with repeated or sustained use, such as holding the arms up.

Not cues: back pain after gardening is a common strain, her coffee intake is moderate, and her heart rate and blood pressure are normal.

Also note what she denies: MG affects only the neuromuscular junction, so sensation is normal (numbness and tingling would suggest a nerve problem) and there is no twitching (fasciculations suggest a motor neuron disease such as ALS). Weakness that is worst on waking would be the opposite of the MG pattern.

Case Study 5 (Myasthenia gravis), screen 2 of 6: Analyze cues



Case Study 5, screen 2: rationale

CORRECT **Score: 4 / 4 points**

0/1 scoring: 1 point for each blank answered correctly.

Rationale:

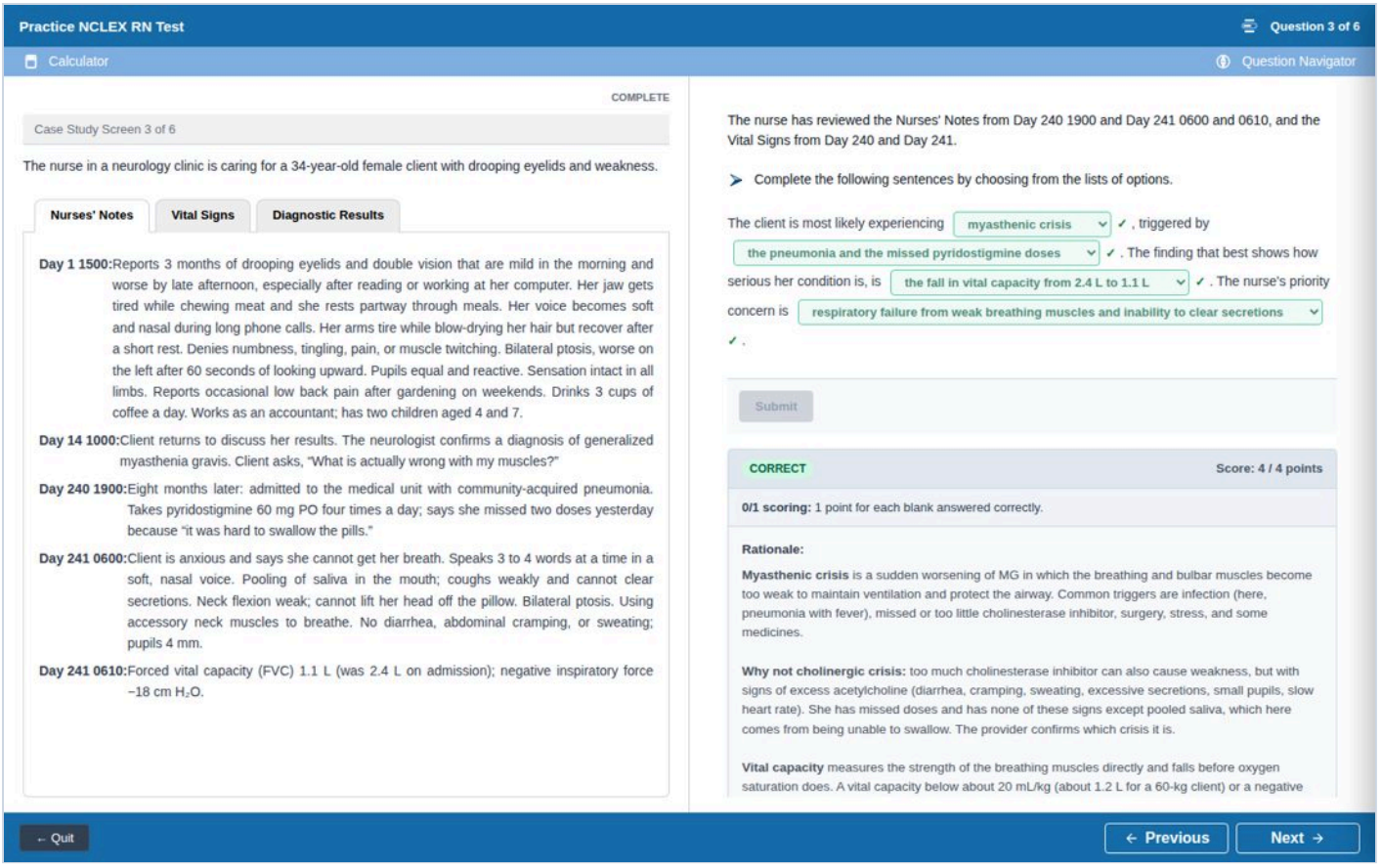
Autoimmune mechanism: MG is an antibody-mediated autoimmune disease. Antibodies against the acetylcholine (ACh) receptor on the muscle side of the neuromuscular junction block the receptors, speed their destruction, and activate complement, which damages the folds of the muscle membrane. Fewer receptors mean a weaker signal to contract. (Motor neuron degeneration is ALS; demyelination of peripheral nerves is Guillain-Barré syndrome; absent dystrophin is Duchenne muscular dystrophy.)

Fatigability: with each repeated nerve impulse, the amount of ACh released normally falls a little. In healthy muscle there are plenty of receptors, so this does not matter; in MG there are too few, so contraction fades. This is also what repetitive nerve stimulation shows: a decreasing response.

Ice pack test: cooling slows acetylcholinesterase, the enzyme that breaks down ACh, so more ACh is available at the receptors and the eyelid lifts. Heat has the opposite effect, which is why hot weather and hot baths worsen MG.

Thymus: most clients with ACh receptor antibodies have an abnormal thymus (hyperplasia, or a thymoma in about 10% to 15%). The thymus is thought to help start and maintain the autoimmune response.

Case Study 5 (Myasthenia gravis), screen 3 of 6: Prioritize hypotheses



Case Study 5, screen 3: rationale

CORRECT **Score: 4 / 4 points**

0/1 scoring: 1 point for each blank answered correctly.

Rationale:

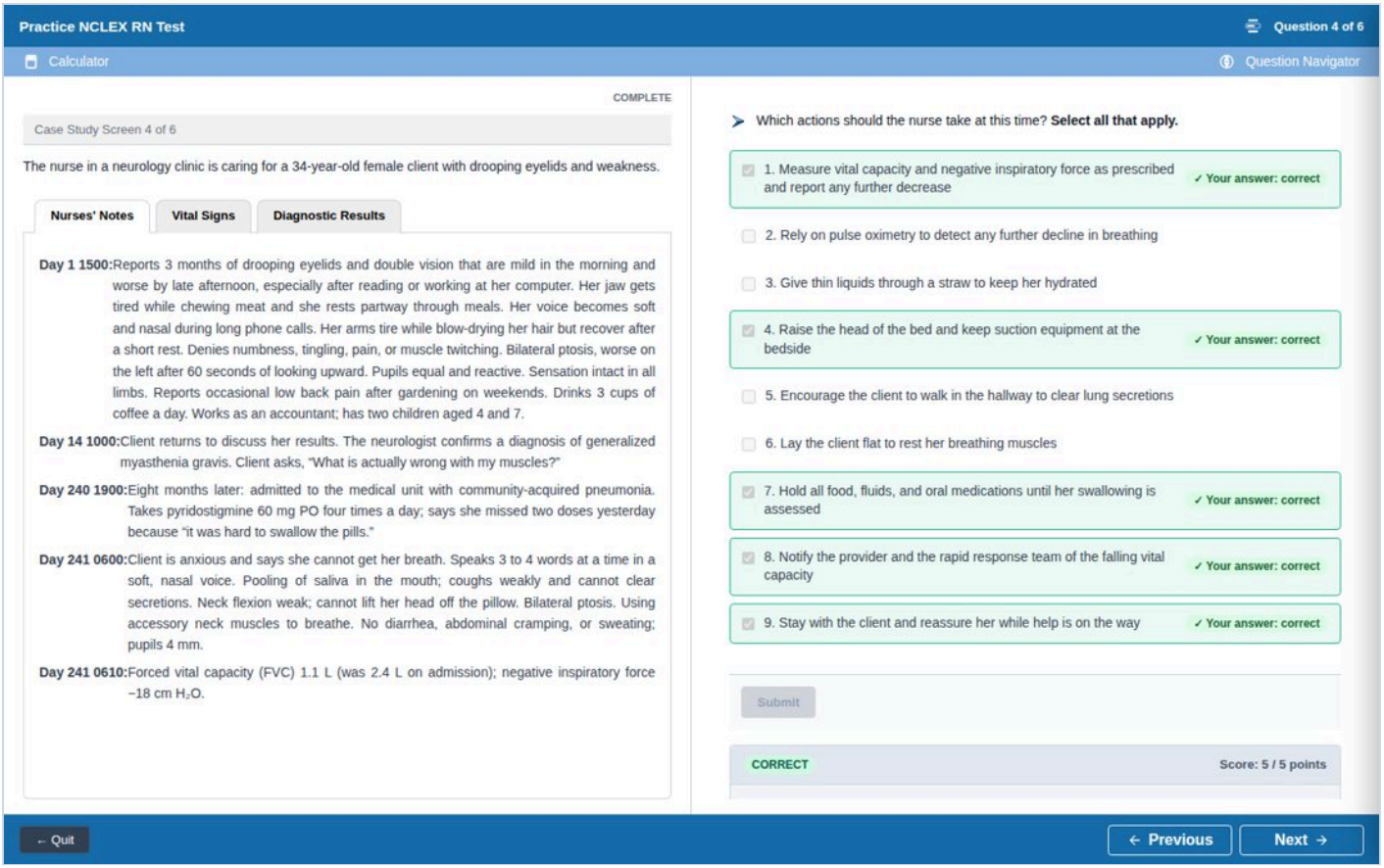
Myasthenic crisis is a sudden worsening of MG in which the breathing and bulbar muscles become too weak to maintain ventilation and protect the airway. Common triggers are infection (here, pneumonia with fever), missed or too little cholinesterase inhibitor, surgery, stress, and some medicines.

Why not cholinergic crisis: too much cholinesterase inhibitor can also cause weakness, but with signs of excess acetylcholine (diarrhea, cramping, sweating, excessive secretions, small pupils, slow heart rate). She has missed doses and has none of these signs except pooled saliva, which here comes from being unable to swallow. The provider confirms which crisis it is.

Vital capacity measures the strength of the breathing muscles directly and falls before oxygen saturation does. A vital capacity below about 20 mL/kg (about 1.2 L for a 60-kg client) or a negative inspiratory force weaker than −20 to −30 cm H_2O signals impending respiratory failure. Pulse oximetry stays reassuring until late because the oxygen she is receiving masks hypoventilation.

Priority: airway and breathing. Weak respiratory muscles plus weak swallowing and coughing (aspiration risk) can lead to respiratory arrest within hours.

Case Study 5 (Myasthenia gravis), screen 4 of 6: Generate solutions



Case Study 5, screen 4: rationale

CORRECT

Score: 5 / 5 points

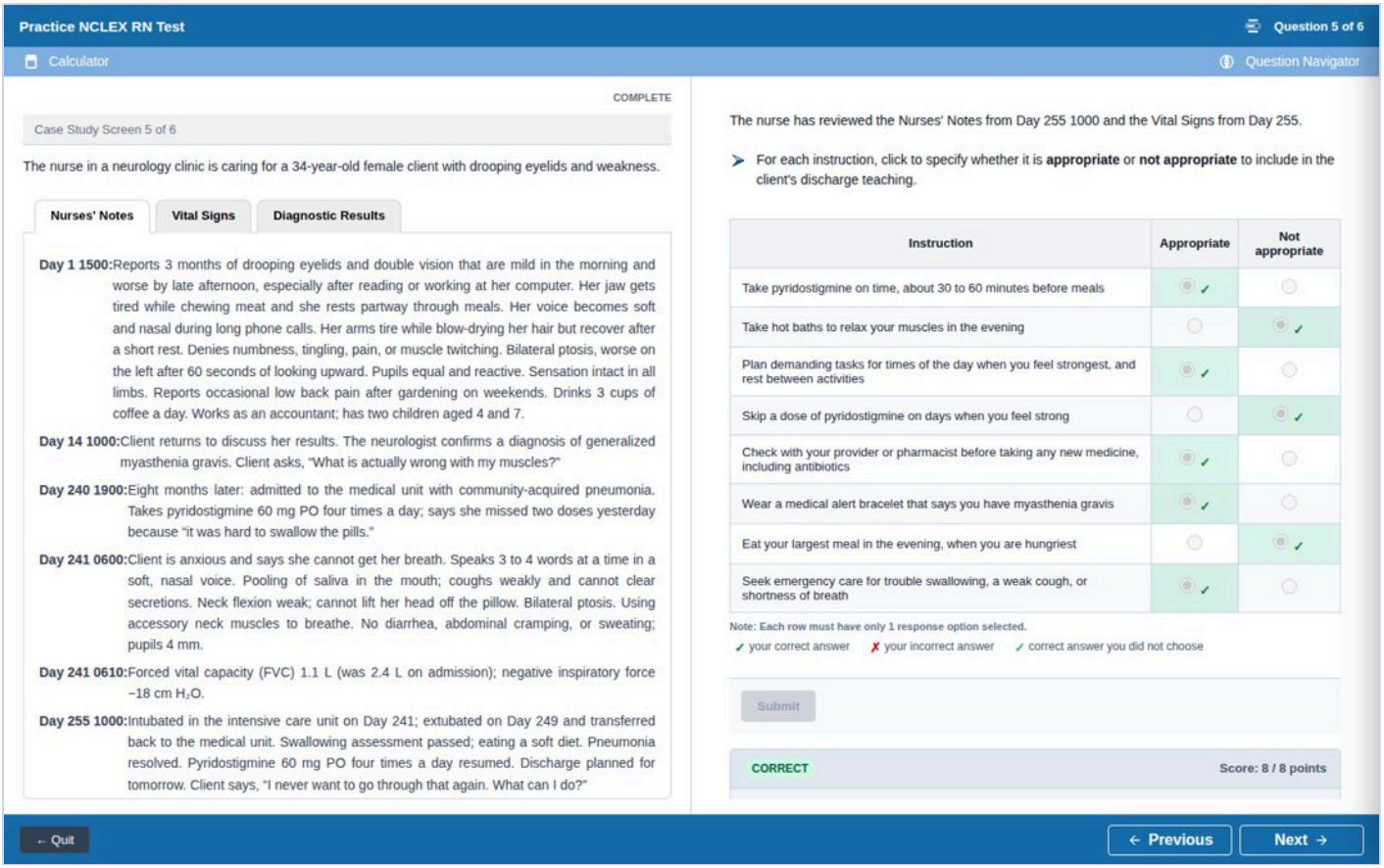
+/− scoring: each correct choice earns 1 point and each incorrect choice takes 1 point away (never below 0). You selected 5 correct (+5) and 0 incorrect (−0) = 5 points.

Rationale:

Correct: a vital capacity this low means she may soon need ventilatory support, so the provider and the rapid response team are called now. Head elevation helps breathing and reduces aspiration, and suction must be ready because she cannot clear her secretions. Nothing is given by mouth while bulbar weakness makes swallowing unsafe; the nurse tells the provider that oral medicines are being held so other routes can be ordered. Serial vital capacity and negative inspiratory force track her breathing muscle strength. Staying with her reduces anxiety and allows rapid response to deterioration.

Incorrect: exertion worsens fatigable weakness and increases oxygen demand. Thin liquids are the hardest to swallow safely and are easily aspirated. Lying flat makes breathing harder and increases aspiration risk. Pulse oximetry detects respiratory failure late, especially when the client is receiving oxygen; vital capacity and the client's work of breathing are better guides.

Case Study 5 (Myasthenia gravis), screen 5 of 6: Take action



Case Study 5, screen 5: rationale

CORRECT

Score: 8 / 8 points

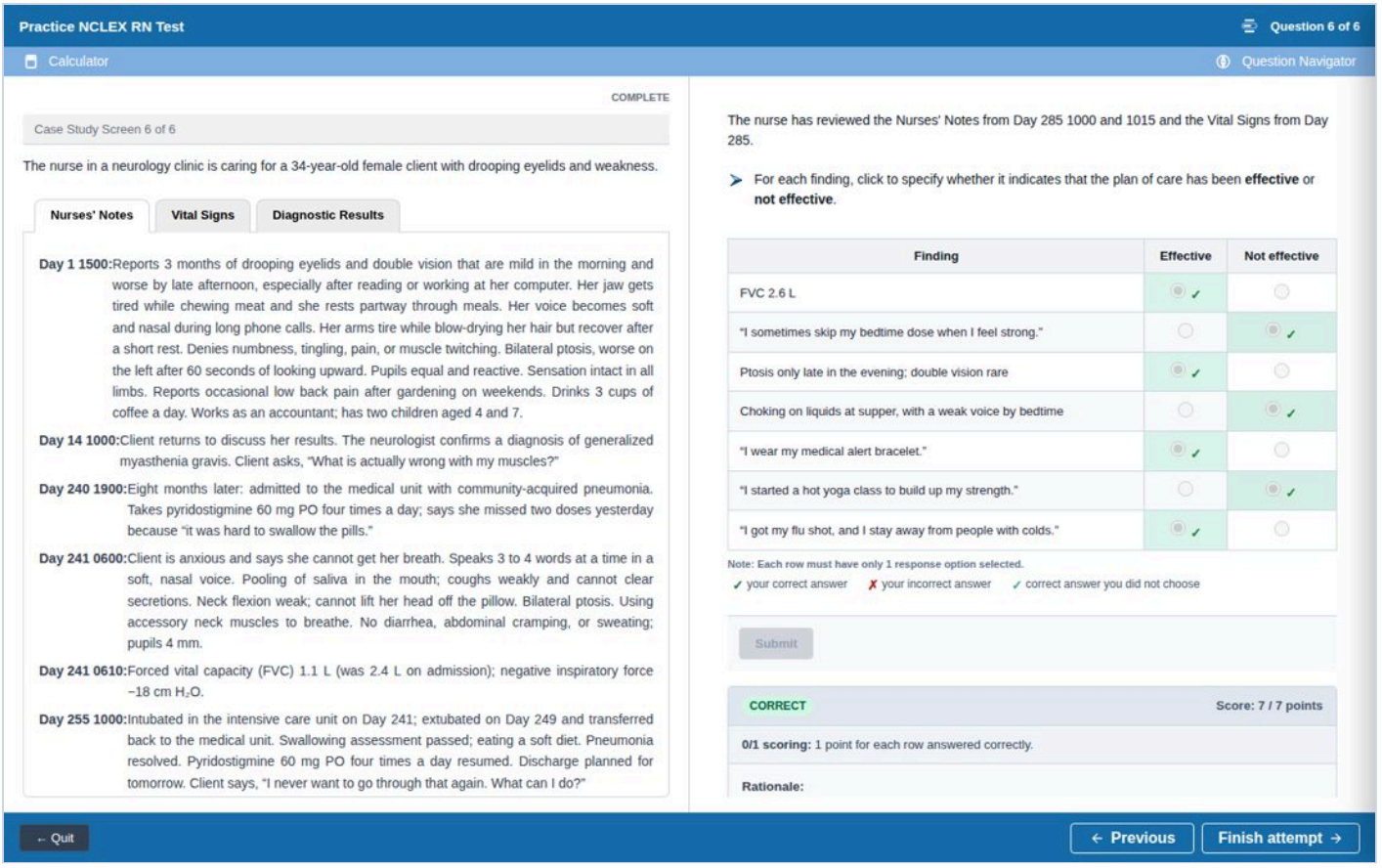
0/1 scoring: 1 point for each row answered correctly.

Rationale:

Appropriate: pyridostigmine (a cholinesterase inhibitor) works for 3 to 4 hours, so doses are taken on time, and before meals so chewing and swallowing are strongest while eating. Energy conservation (doing demanding tasks when strength is best, resting between activities) works with the fatigable pattern. Many medicines can worsen MG (for example some antibiotics such as fluoroquinolones and aminoglycosides, magnesium, and beta-blockers), so new medicines are checked first. A medical alert bracelet informs emergency staff. Difficulty swallowing, a weak cough or breathlessness can signal a crisis and need emergency care.

Not appropriate: heat worsens neuromuscular transmission (hot baths, saunas, hot weather). Missing doses can trigger a crisis, as it did before. Fatigue is worst in the evening, so large evening meals raise the risk of choking; smaller meals at times of best strength are safer.

Case Study 5 (Myasthenia gravis), screen 6 of 6: Evaluate outcomes



Case Study 5, screen 6: rationale

CORRECT

Score: 7 / 7 points

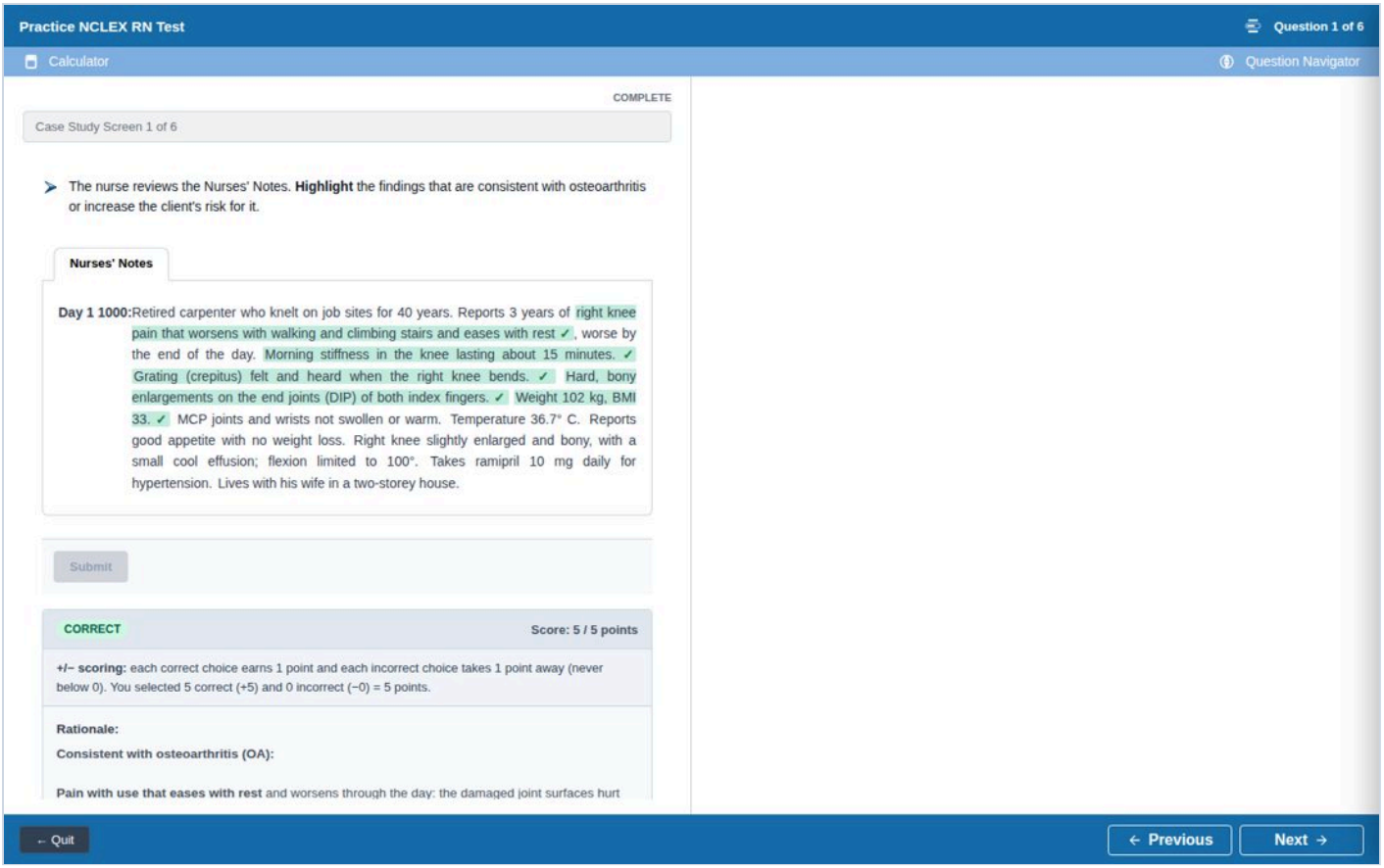
0/1 scoring: 1 point for each row answered correctly.

Rationale:

Effective: her vital capacity has recovered to a safe level, her ocular symptoms are mild and limited to the end of the day (expected in MG), and she has taken steps that reduce the risk of another crisis (medical alert bracelet, influenza vaccination with an inactivated vaccine, avoiding infection).

Not effective: skipping doses and choking with a weak voice in the evening show that bulbar weakness is not controlled at the end of the day; this puts her at risk of aspiration and another crisis. The nurse reinforces taking every dose on time and reports the evening choking to the provider. Hot yoga exposes her to heat, which worsens neuromuscular transmission; gentle exercise in a cool environment, paced with rest, is safer.

Case Study 6 (Osteoarthritis), screen 1 of 6: Recognize cues



Case Study 6, screen 1: rationale

CORRECT **Score: 5 / 5 points**

+/− scoring: each correct choice earns 1 point and each incorrect choice takes 1 point away (never below 0). You selected 5 correct (+5) and 0 incorrect (−0) = 5 points.

Rationale:

Consistent with osteoarthritis (OA):

Pain with use that eases with rest and worsens through the day: the damaged joint surfaces hurt under load. In late disease pain may also occur at rest and at night.

Short morning stiffness (usually under 30 minutes) after rest, which loosens with movement.

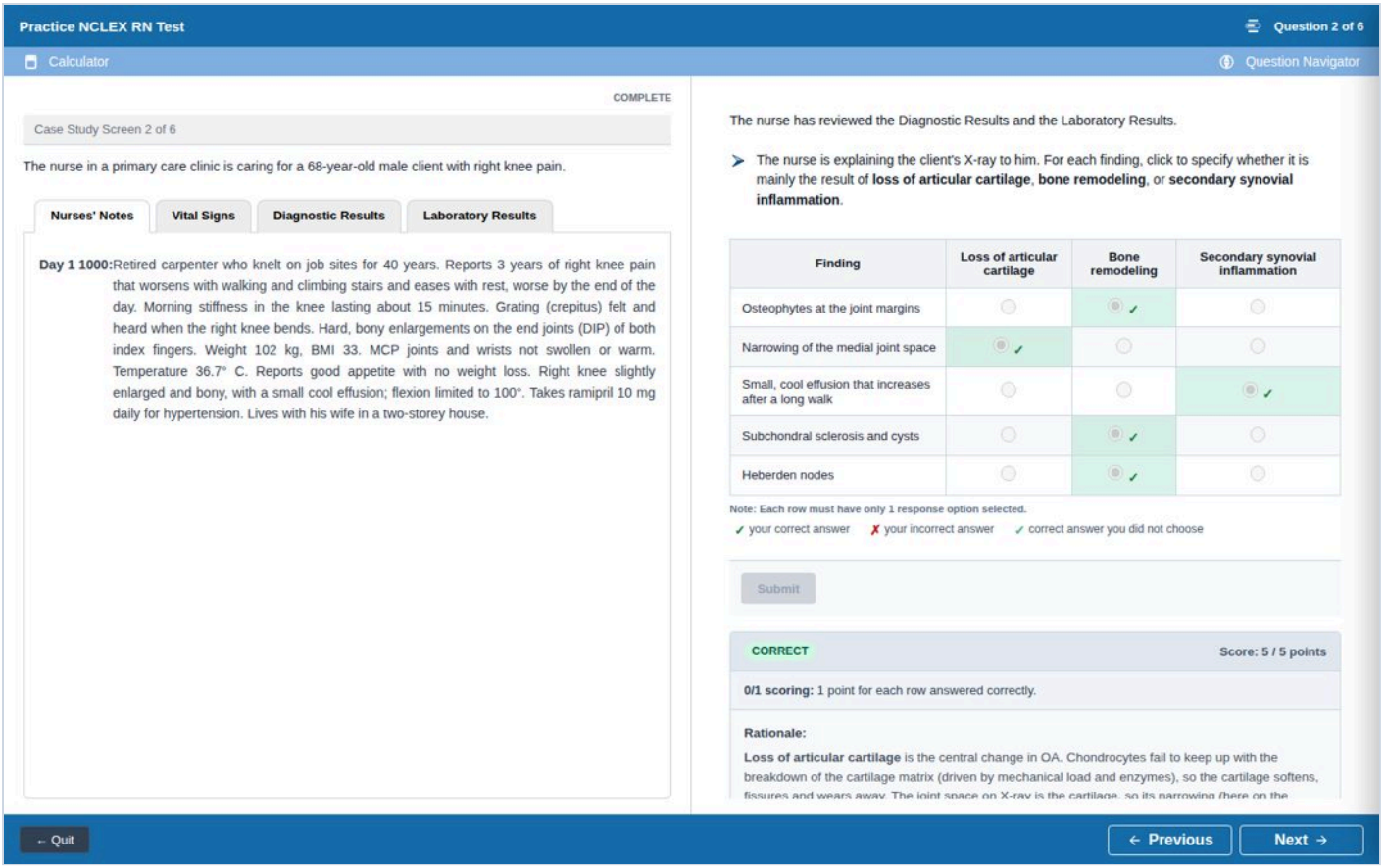
Crepitus: grating from roughened cartilage and bone surfaces moving against each other.

Heberden nodes: bony enlargements (osteophytes) of the distal interphalangeal joints; Bouchard nodes affect the proximal interphalangeal joints.

Risk factors: obesity (more load on weight-bearing joints, plus inflammatory substances from fat tissue) and years of repeated kneeling or joint stress. Age and previous joint injury are other major risk factors.

Not OA cues: unswollen MCP joints and wrists, a normal temperature and no weight loss argue against an inflammatory, systemic arthritis such as rheumatoid arthritis. Ramipril treats his hypertension.

Case Study 6 (Osteoarthritis), screen 2 of 6: Analyze cues



Case Study 6, screen 2: rationale

CORRECT

Score: 5 / 5 points

0/1 scoring: 1 point for each row answered correctly.

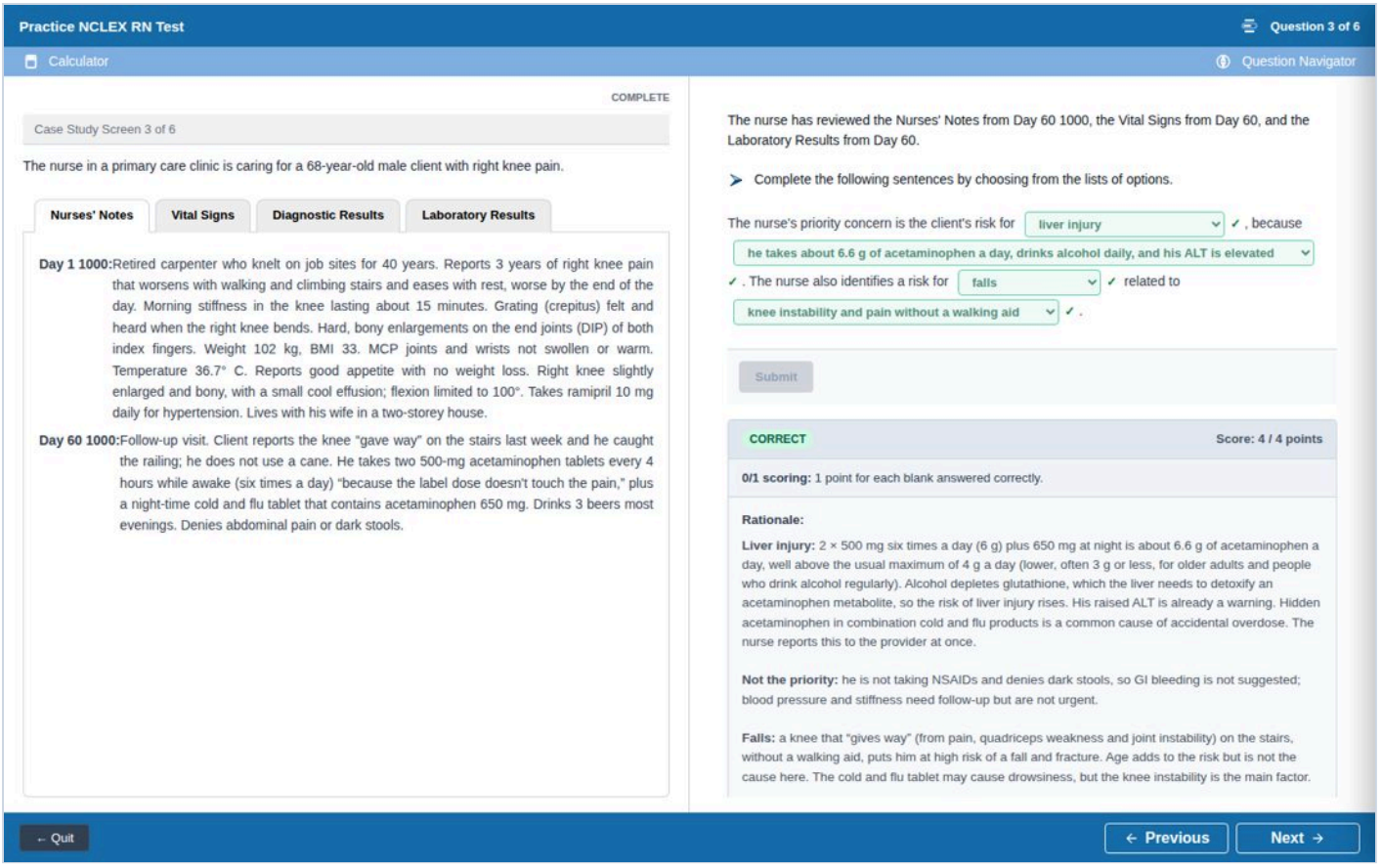
Rationale:

Loss of articular cartilage is the central change in OA. Chondrocytes fail to keep up with the breakdown of the cartilage matrix (driven by mechanical load and enzymes), so the cartilage softens, fissures and wears away. The joint space on X-ray is the cartilage, so its narrowing (here on the medial side, which carries the most load) shows cartilage loss.

Bone remodeling: as cartilage is lost, the bone underneath takes the load. It thickens and becomes dense (subchondral sclerosis), and joint fluid forced into cracks forms cysts. At the joint margins, new bone grows as spurs (osteophytes), which limit motion and, in the fingers, form Heberden and Bouchard nodes.

Secondary synovial inflammation: fragments of cartilage irritate the synovium, which produces a mild, cool effusion after activity. This inflammation is local and mild, so the ESR and CRP are normal and the RF is negative, unlike in rheumatoid arthritis. The normal uric acid argues against gout.

Case Study 6 (Osteoarthritis), screen 3 of 6: Prioritize hypotheses



Case Study 6, screen 3: rationale

CORRECT **Score: 4 / 4 points**

0/1 scoring: 1 point for each blank answered correctly.

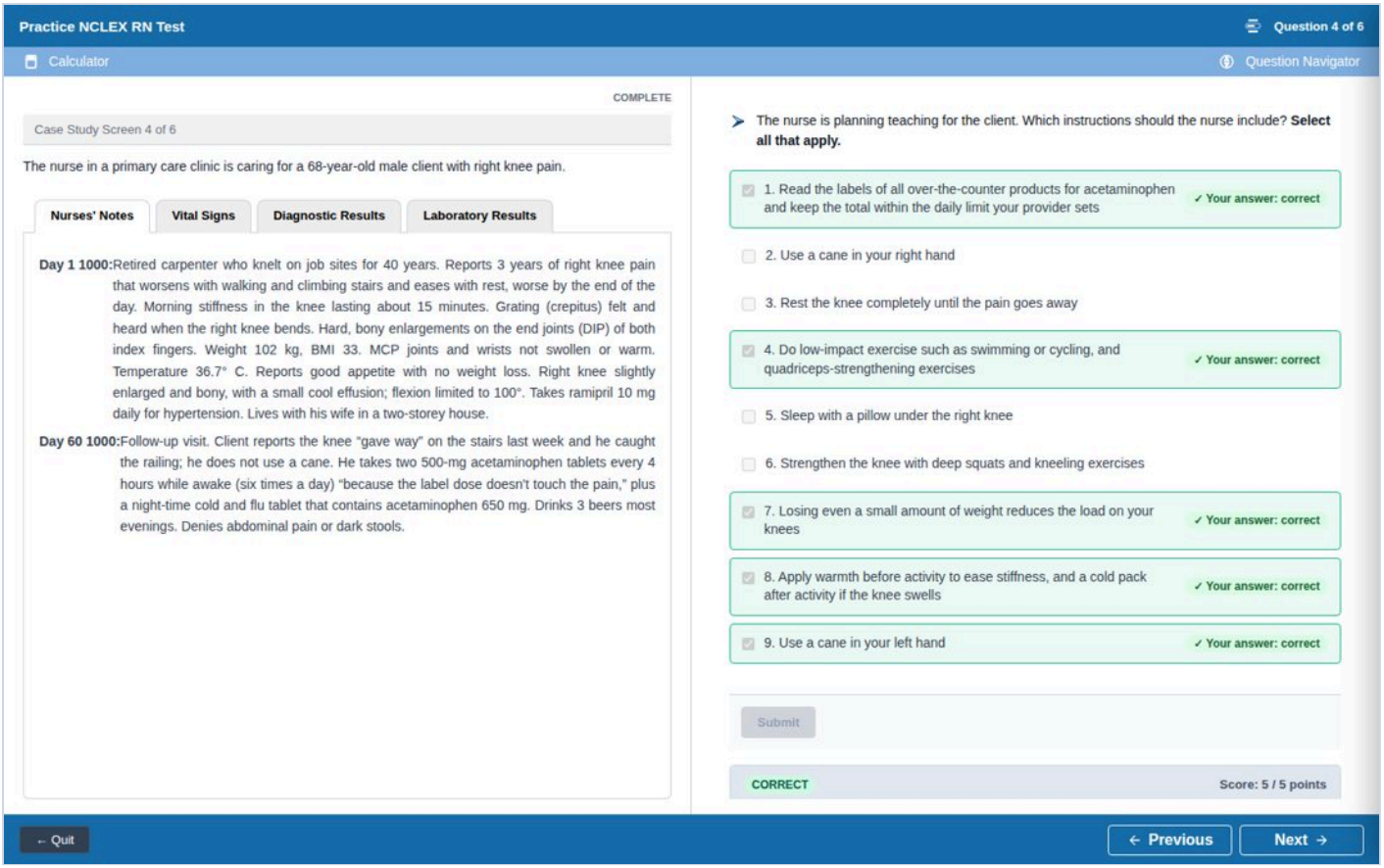
Rationale:

Liver injury: 2 × 500 mg six times a day (6 g) plus 650 mg at night is about 6.6 g of acetaminophen a day, well above the usual maximum of 4 g a day (lower, often 3 g or less, for older adults and people who drink alcohol regularly). Alcohol depletes glutathione, which the liver needs to detoxify an acetaminophen metabolite, so the risk of liver injury rises. His raised ALT is already a warning. Hidden acetaminophen in combination cold and flu products is a common cause of accidental overdose. The nurse reports this to the provider at once.

Not the priority: he is not taking NSAIDs and denies dark stools, so GI bleeding is not suggested; blood pressure and stiffness need follow-up but are not urgent.

Falls: a knee that "gives way" (from pain, quadriceps weakness and joint instability) on the stairs, without a walking aid, puts him at high risk of a fall and fracture. Age adds to the risk but is not the cause here. The cold and flu tablet may cause drowsiness, but the knee instability is the main factor.

Case Study 6 (Osteoarthritis), screen 4 of 6: Generate solutions



Case Study 6, screen 4: rationale

CORRECT

Score: 5 / 5 points

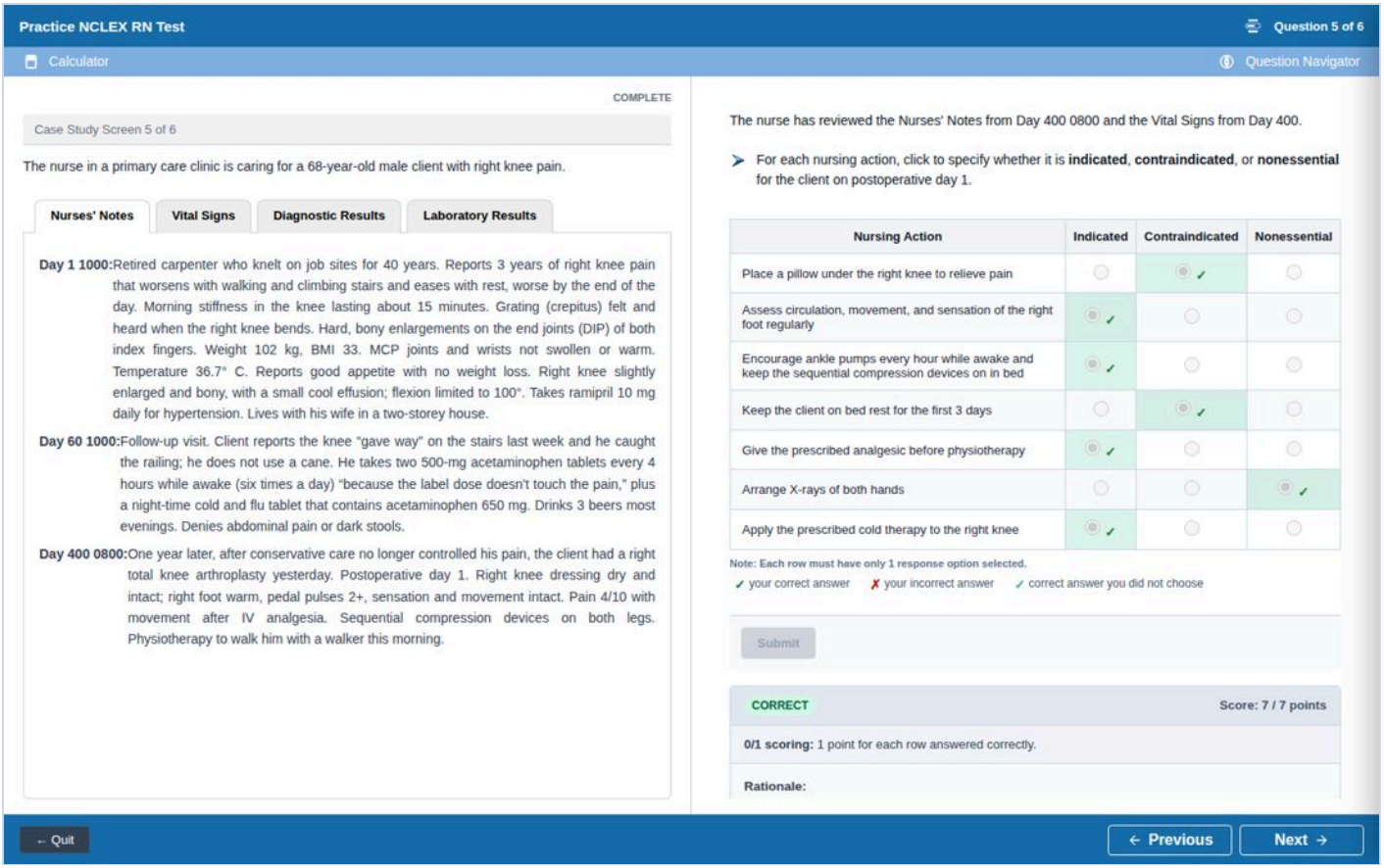
+/− scoring: each correct choice earns 1 point and each incorrect choice takes 1 point away (never below 0). You selected 5 correct (+5) and 0 incorrect (−0) = 5 points.

Rationale:

Correct: checking labels prevents hidden acetaminophen overdose. A cane is held in the hand opposite the affected knee and moved forward with it, which shifts weight off the painful knee and widens the base of support. Heat relaxes muscles and eases stiffness before activity; cold reduces swelling and pain after activity. Low-impact aerobic exercise and quadriceps strengthening stabilize the knee and reduce pain without overloading it. Every kilogram of weight lost removes about 4 kg of load from the knee with each step.

Incorrect: a cane on the same side as the bad knee does not unload it. Complete rest causes stiffness, muscle weakness and further loss of function. Deep squats and kneeling load the damaged cartilage heavily. A pillow under the knee is comfortable but encourages a flexion contracture.

Case Study 6 (Osteoarthritis), screen 5 of 6: Take action



Case Study 6, screen 5: rationale

CORRECT

Score: 7 / 7 points

0/1 scoring: 1 point for each row answered correctly.

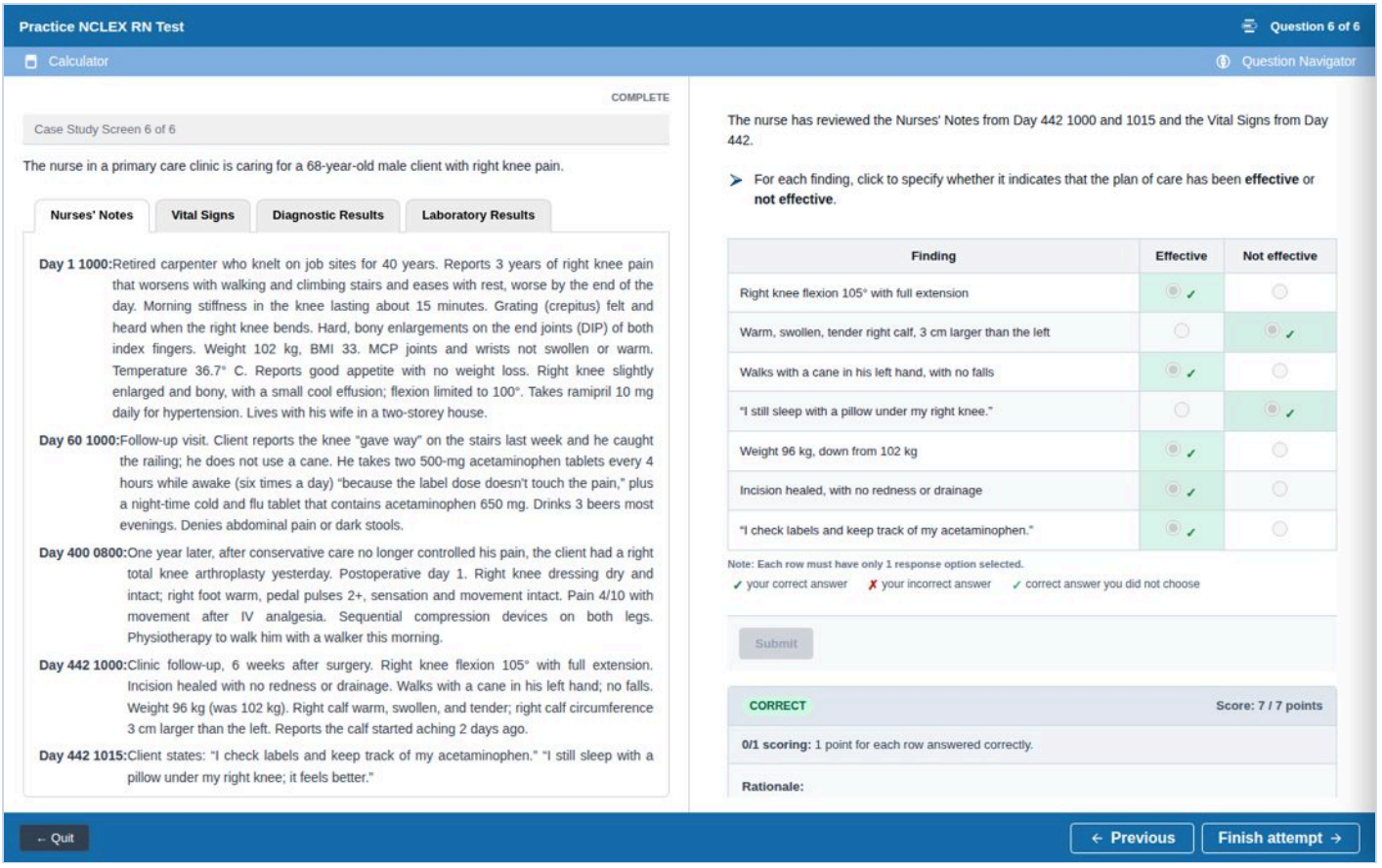
Rationale:

Indicated: neurovascular checks detect nerve or vascular injury and compartment problems early. Ankle pumps and sequential compression devices promote venous return and help prevent deep vein thrombosis, a major risk after knee replacement. Pain relief before physiotherapy lets him take part in early walking and range-of-motion exercise. Cold therapy reduces swelling and pain.

Contraindicated: a pillow under the knee holds it flexed and can cause a flexion contracture; the knee is kept straight when resting (a pillow may go under the calf and heel if the leg is elevated). Bed rest increases the risk of blood clots, pneumonia and stiffness; walking usually starts on the day of surgery or the day after.

Nonessential: hand X-rays do not change his postoperative care.

Case Study 6 (Osteoarthritis), screen 6 of 6: Evaluate outcomes



Case Study 6, screen 6: rationale

CORRECT

Score: 7 / 7 points

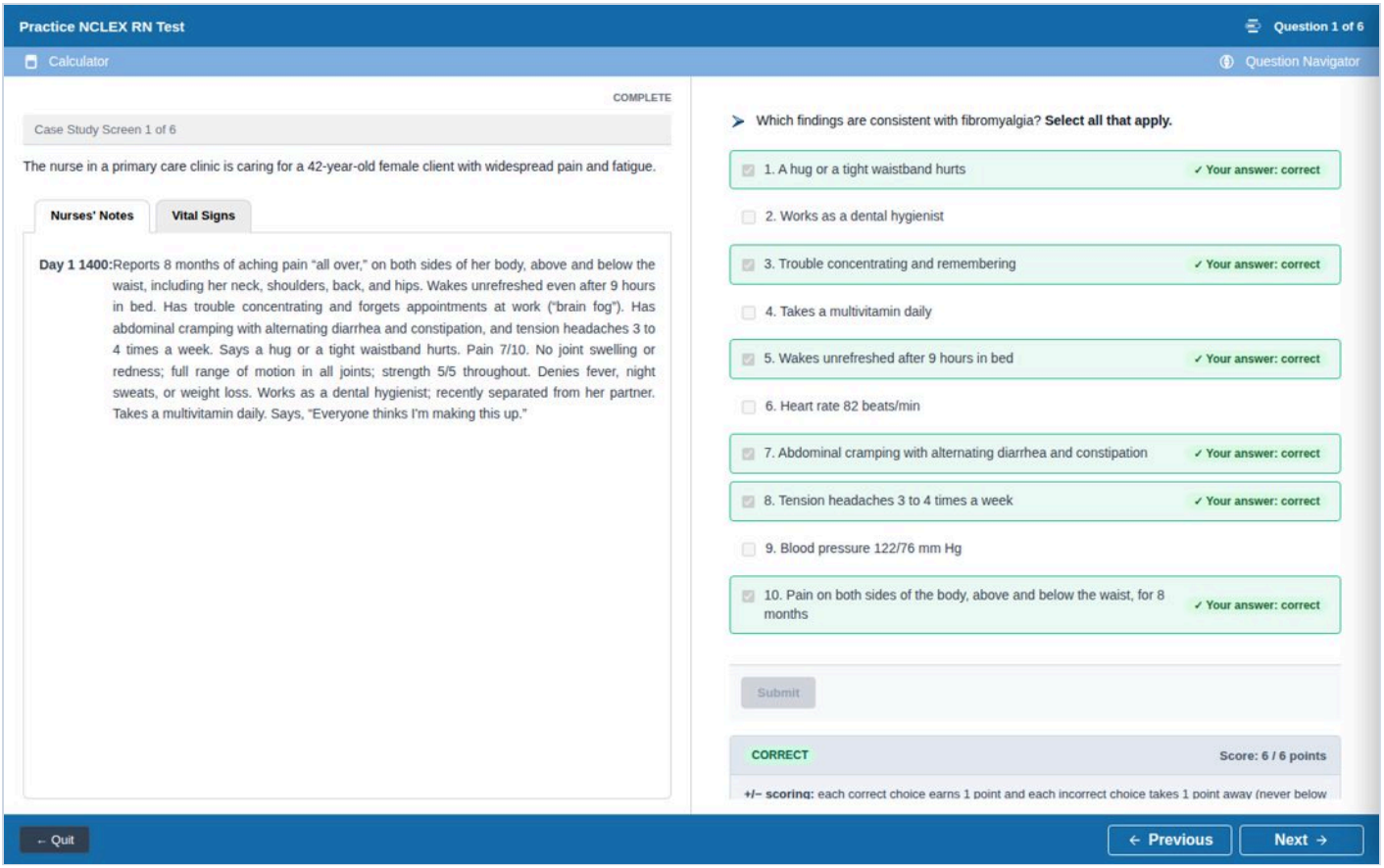
0/1 scoring: 1 point for each row answered correctly.

Rationale:

Effective: good range of motion (about 90° to 110° of flexion is a usual goal at this stage), a healed incision, safe use of the cane on the correct side with no falls, weight loss, and safe use of acetaminophen.

Not effective: a warm, swollen, tender calf that is larger than the other suggests a deep vein thrombosis, which can lead to a pulmonary embolism. The nurse does not massage the calf, has the client rest, and reports the finding to the provider immediately. Sleeping with a pillow under the knee risks a flexion contracture and loss of the extension gained in therapy; the nurse reteaches keeping the knee straight.

Case Study 7 (Fibromyalgia), screen 1 of 6: Recognize cues



Case Study 7, screen 1: rationale

CORRECT

Score: 6 / 6 points

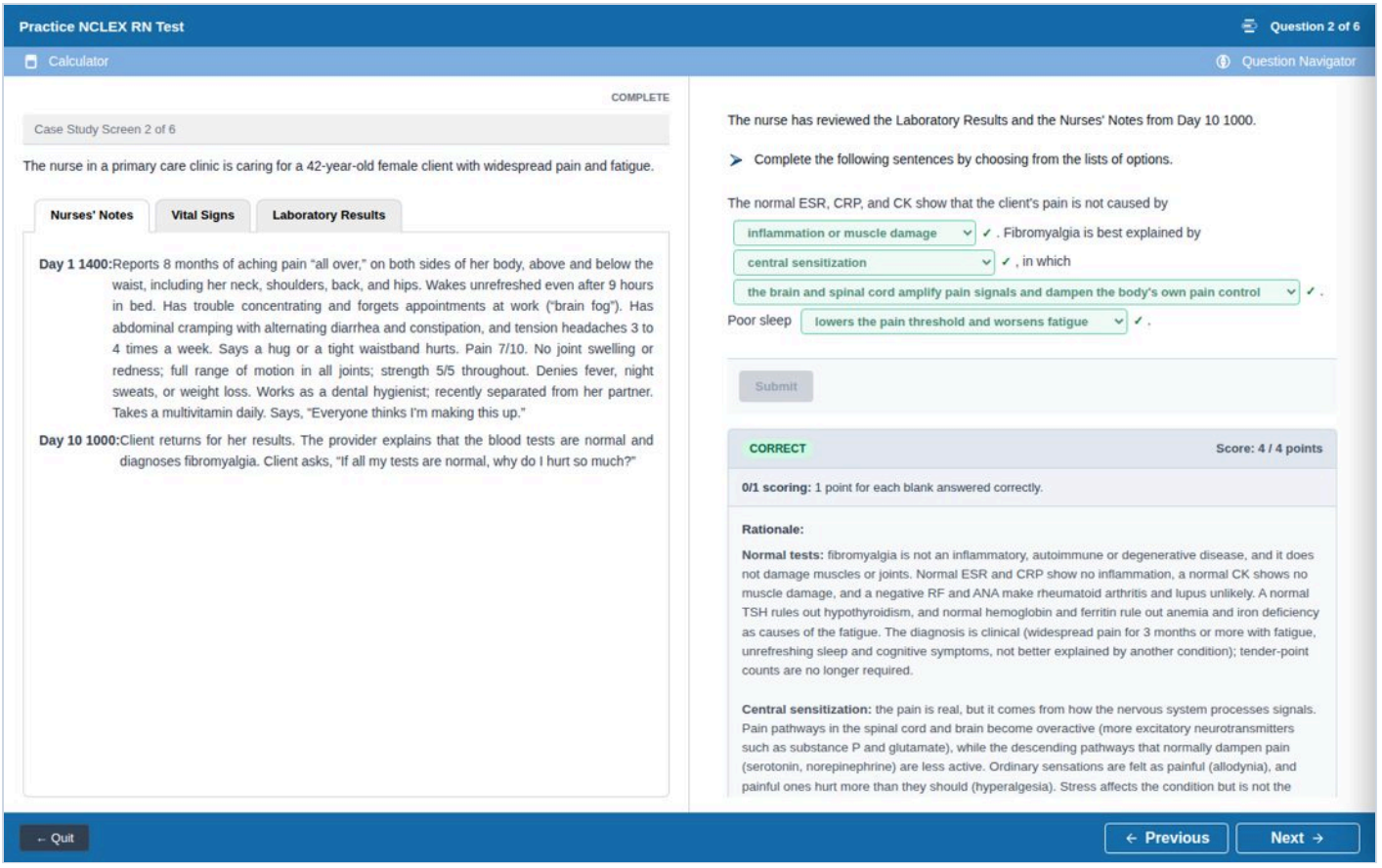
+/− scoring: each correct choice earns 1 point and each incorrect choice takes 1 point away (never below 0). You selected 6 correct (+6) and 0 incorrect (−0) = 6 points.

Rationale:

Consistent with fibromyalgia: chronic (3 months or more) **widespread pain** on both sides of the body and above and below the waist is the core feature. It is accompanied by **unrefreshing sleep**, **fatigue**, and **cognitive symptoms** ("fibro fog"). Pain from touch that should not hurt (**allodynia**, such as a hug or a waistband) reflects an oversensitive pain system. Other functional conditions are common alongside it: **irritable bowel syndrome**, **tension headaches** and migraines, temporomandibular disorder, and anxiety or depression.

Not cues: a multivitamin, a normal heart rate and blood pressure, and her job. Note also what is absent: no joint swelling, normal strength and range of motion, and no fever or weight loss. Fibromyalgia does not damage joints or muscles, and swelling, fever, weight loss or true muscle weakness would suggest another diagnosis.

Case Study 7 (Fibromyalgia), screen 2 of 6: Analyze cues



Case Study 7, screen 2: rationale

CORRECT

Score: 4 / 4 points

0/1 scoring: 1 point for each blank answered correctly.

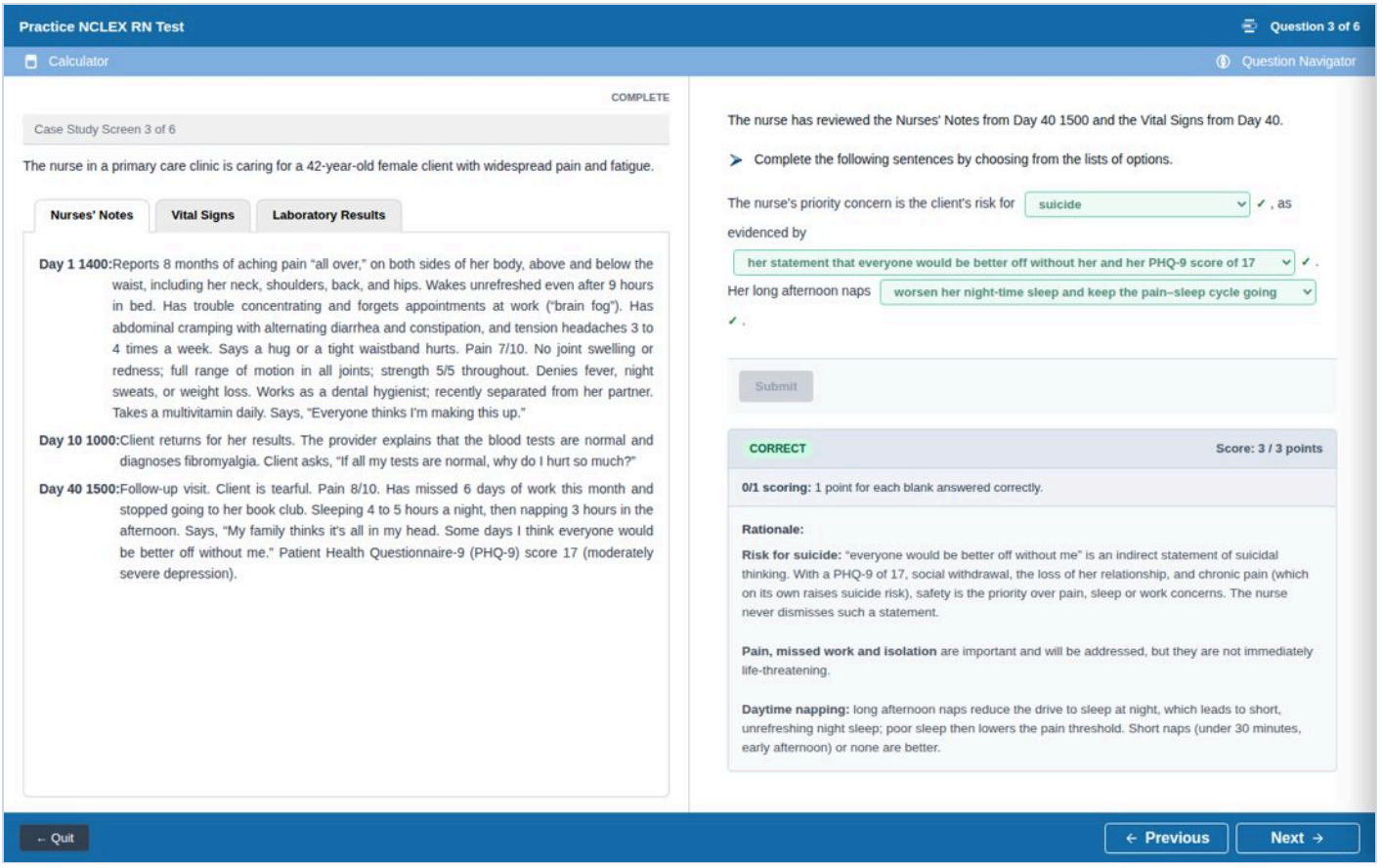
Rationale:

Normal tests: fibromyalgia is not an inflammatory, autoimmune or degenerative disease, and it does not damage muscles or joints. Normal ESR and CRP show no inflammation, a normal CK shows no muscle damage, and a negative RF and ANA make rheumatoid arthritis and lupus unlikely. A normal TSH rules out hypothyroidism, and normal hemoglobin and ferritin rule out anemia and iron deficiency as causes of the fatigue. The diagnosis is clinical (widespread pain for 3 months or more with fatigue, unrefreshing sleep and cognitive symptoms, not better explained by another condition); tender-point counts are no longer required.

Central sensitization: the pain is real, but it comes from how the nervous system processes signals. Pain pathways in the spinal cord and brain become overactive (more excitatory neurotransmitters such as substance P and glutamate), while the descending pathways that normally dampen pain (serotonin, norepinephrine) are less active. Ordinary sensations are felt as painful (allodynia), and painful ones hurt more than they should (hyperalgesia). Stress affects the condition but is not the cause.

Sleep: people with fibromyalgia have less deep, restorative sleep. Poor sleep lowers the pain threshold and worsens fatigue and concentration, and more pain disturbs sleep further, creating a cycle that nursing care aims to break.

Case Study 7 (Fibromyalgia), screen 3 of 6: Prioritize hypotheses



Case Study 7, screen 3: rationale

CORRECT **Score: 3 / 3 points**

0/1 scoring: 1 point for each blank answered correctly.

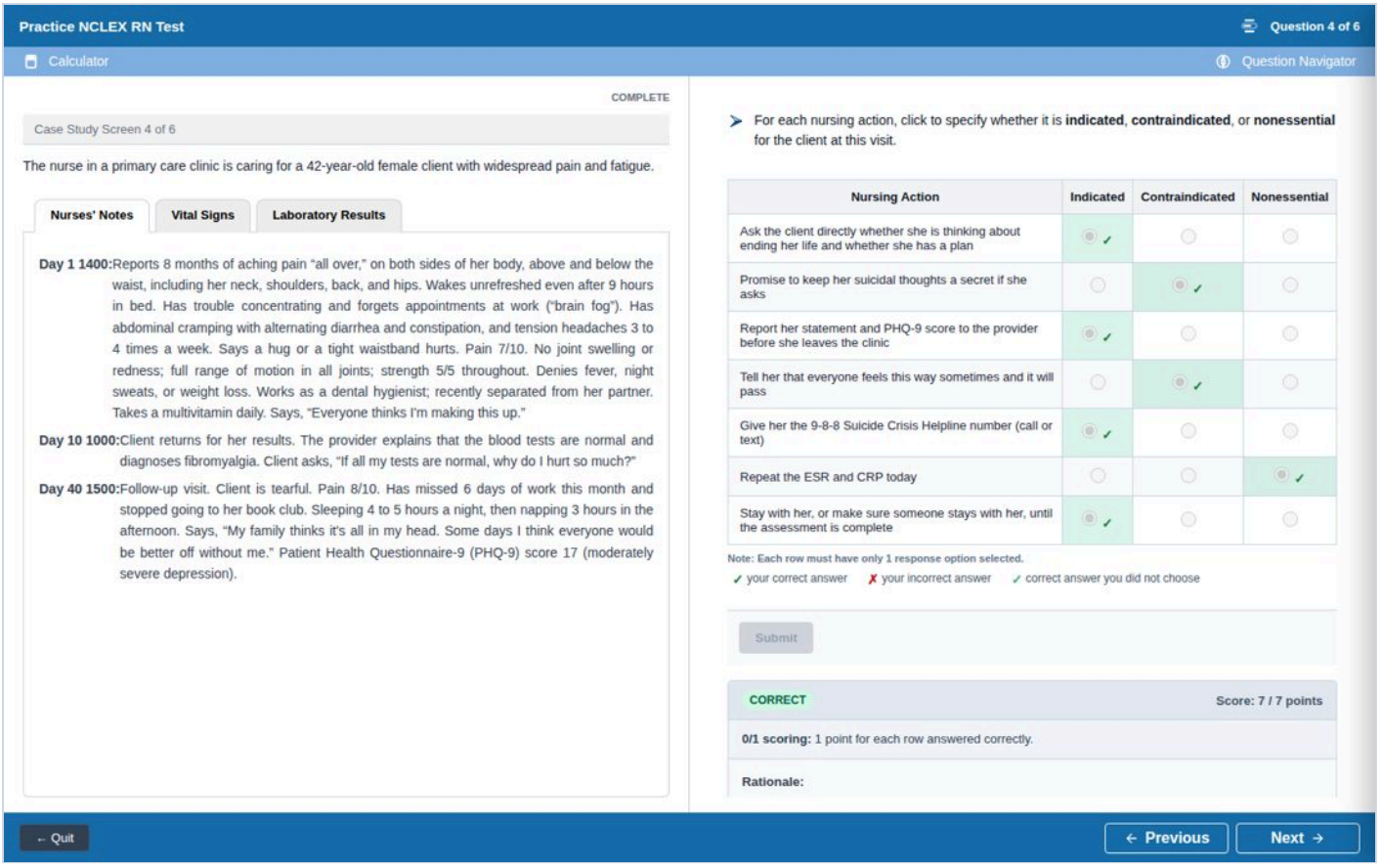
Rationale:

Risk for suicide: "everyone would be better off without me" is an indirect statement of suicidal thinking. With a PHQ-9 of 17, social withdrawal, the loss of her relationship, and chronic pain (which on its own raises suicide risk), safety is the priority over pain, sleep or work concerns. The nurse never dismisses such a statement.

Pain, missed work and isolation are important and will be addressed, but they are not immediately life-threatening.

Daytime napping: long afternoon naps reduce the drive to sleep at night, which leads to short, unrefreshing night sleep; poor sleep then lowers the pain threshold. Short naps (under 30 minutes, early afternoon) or none are better.

Case Study 7 (Fibromyalgia), screen 4 of 6: Generate solutions



Case Study 7, screen 4: rationale

CORRECT

Score: 7 / 7 points

0/1 scoring: 1 point for each row answered correctly.

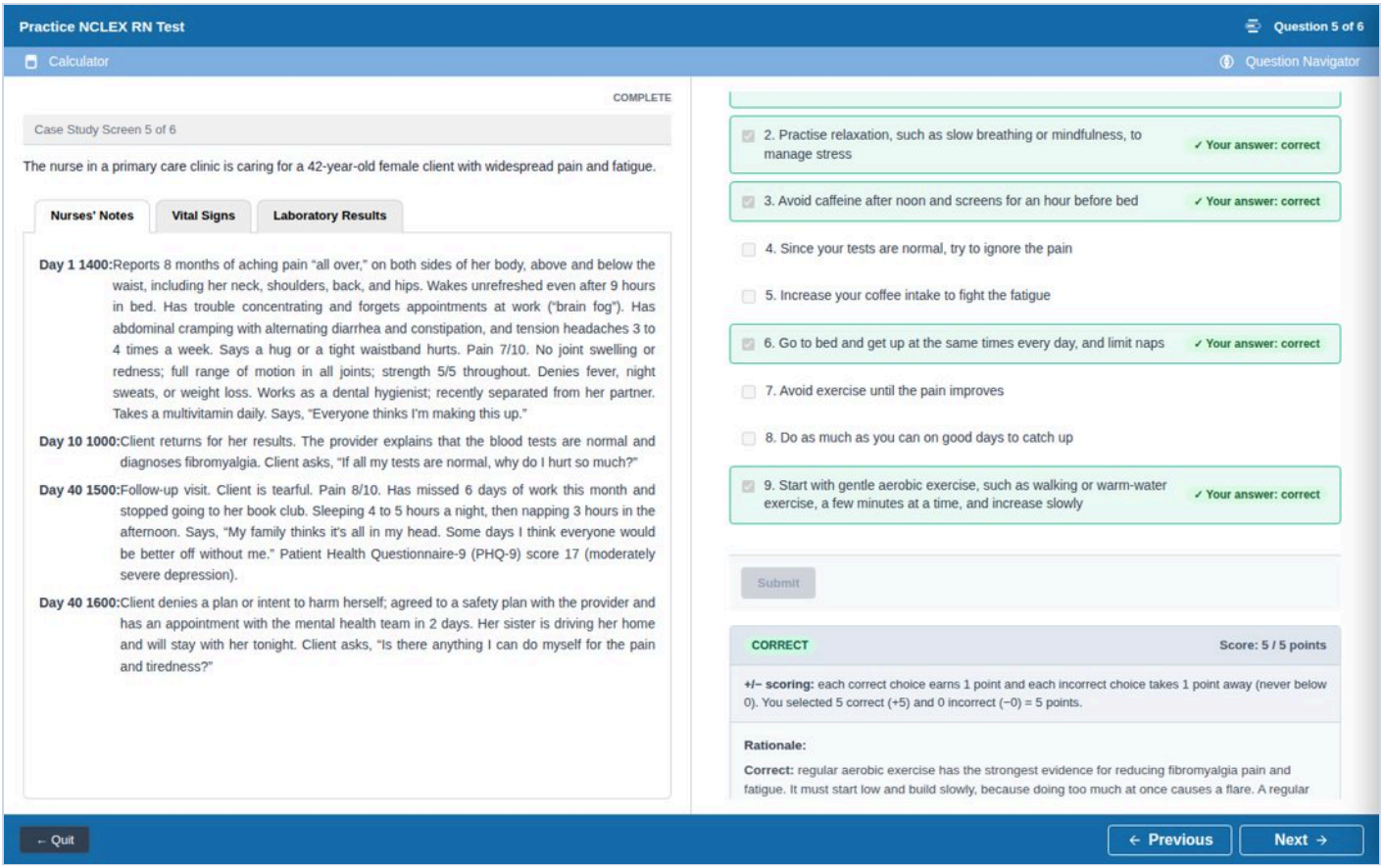
Rationale:

Indicated: asking directly about suicidal thoughts, a plan and the means does not plant the idea; it opens the conversation and is the only way to judge the level of risk. The provider must know before she leaves so that a safety plan and mental health follow-up are arranged. Canada's 9-8-8 Suicide Crisis Helpline is available 24/7 by phone or text. A client who may be at risk is not left alone until the risk has been assessed.

Contraindicated: the nurse cannot promise secrecy about a risk to the client's life; the information must be shared with the team. Saying "everyone feels this way" dismisses her distress and closes the conversation.

Nonessential: her inflammatory markers were normal and repeating them does not address her current risk.

Case Study 7 (Fibromyalgia), screen 5 of 6: Take action



Case Study 7, screen 5: rationale

CORRECT

Score: 5 / 5 points

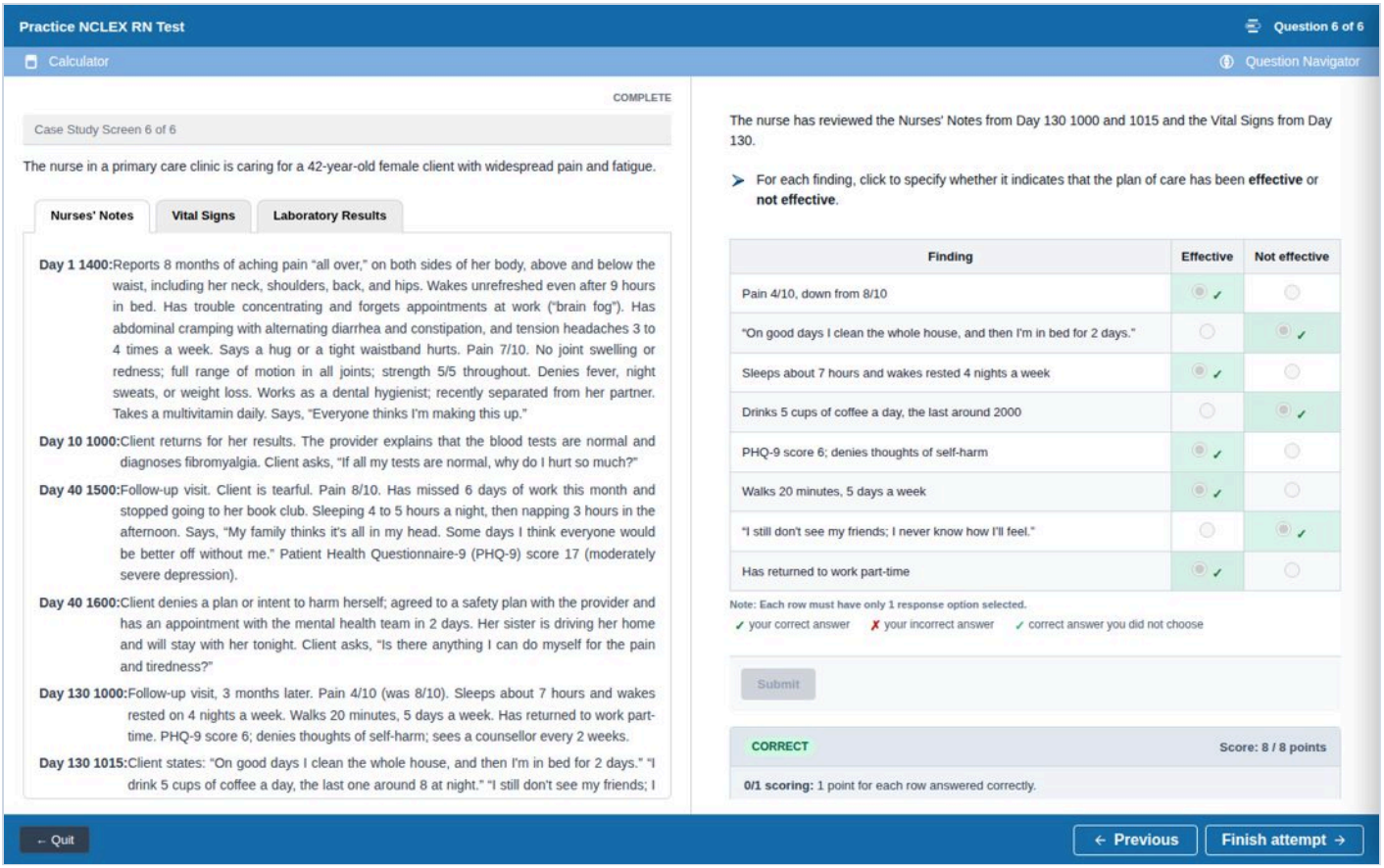
+/− scoring: each correct choice earns 1 point and each incorrect choice takes 1 point away (never below 0). You selected 5 correct (+5) and 0 incorrect (−0) = 5 points.

Rationale:

Correct: regular aerobic exercise has the strongest evidence for reducing fibromyalgia pain and fatigue. It must start low and build slowly, because doing too much at once causes a flare. A regular sleep schedule, limited naps, and avoiding caffeine and screens before bed improve sleep quality (sleep hygiene). Pacing prevents the “boom-and-bust” pattern of overdoing it on good days and crashing afterward. Relaxation and stress management calm the overactive stress and pain systems.

Incorrect: avoiding exercise leads to deconditioning and more pain. Catching up on good days triggers flares. Extra caffeine disrupts sleep. Telling her to ignore the pain dismisses a real condition and damages trust; normal tests mean no tissue damage, not that the pain is imaginary.

Case Study 7 (Fibromyalgia), screen 6 of 6: Evaluate outcomes



Case Study 7, screen 6: rationale

CORRECT **Score: 8 / 8 points**

0/1 scoring: 1 point for each row answered correctly.

Rationale:

Effective: less pain, better sleep, regular aerobic exercise, a return to work, and a much lower depression score with no suicidal thoughts show that the plan is working. Fibromyalgia is chronic; the goal is better function and quality of life, not a cure.

Not effective: the boom-and-bust pattern shows she is not yet pacing herself. Late, heavy caffeine intake undermines sleep. Avoiding friends increases isolation and the risk of depression. The nurse reinforces pacing and sleep hygiene, and explores ways to keep social contact (shorter visits, plans that can be changed).